



**University of
Nottingham**

UK | CHINA | MALAYSIA

COMP1047: Systems and Architecture

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AY2024-25, Spring Semester

Week 6

**Computer Networks Part. Fundamental
Concepts**



Introduction

- Most of the slides are based on the Books

1. Computer Networking: A Top-Down Approach

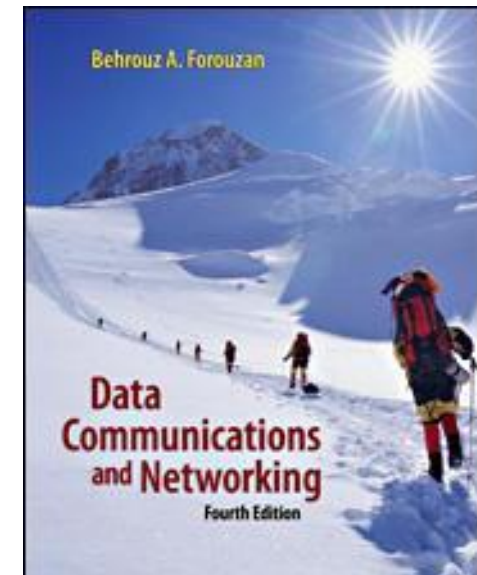
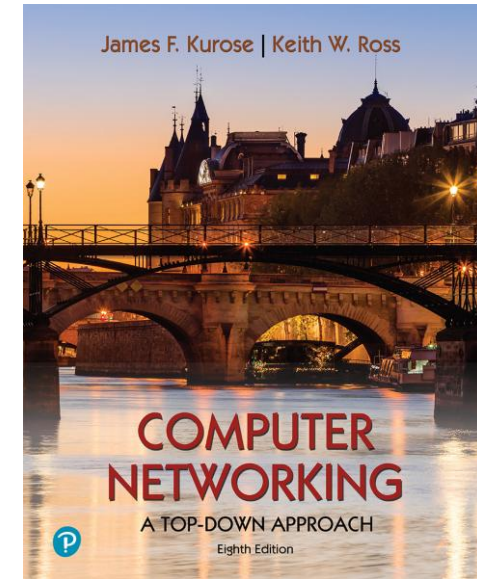
8th edition by Jim Kurose, Keith Ross

and

2. Data Communication and Networking

4th edition by Behrouz A. Forouzan

(Some of these slides are based on Dr. Fazl Ullah and Prof. Ying Weng slides)





Learning Outcomes:

- Learn the basic of networks and networking terminologies
- Understand the types of networks and how they are designed
- The basics of Internet



Overview/roadmap:

- Data and Communication
- Network and its types
- Topologies
- The Internet
- Protocol
- Network Edge (end systems)
- Access Networks and Physical Media
- Network core

Our goal here is to paint a broad picture and set the context for the rest of the lectures, to see the forest through the trees.

We'll cover a lot of ground in this introductory chapter and discuss a lot of the pieces of a computer network, without losing sight of the big picture.

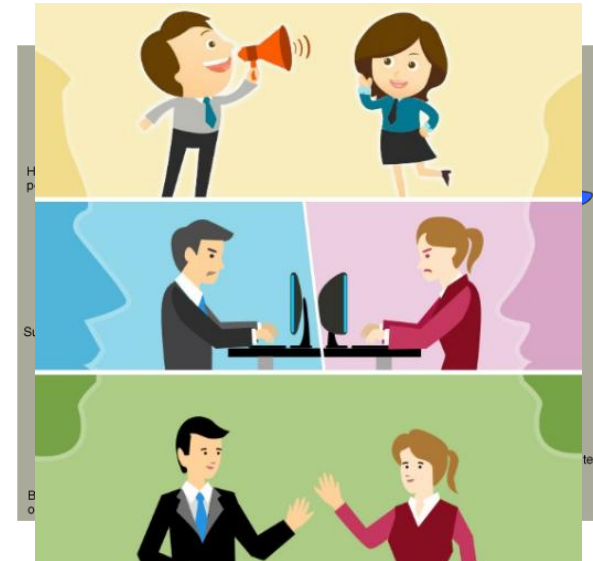
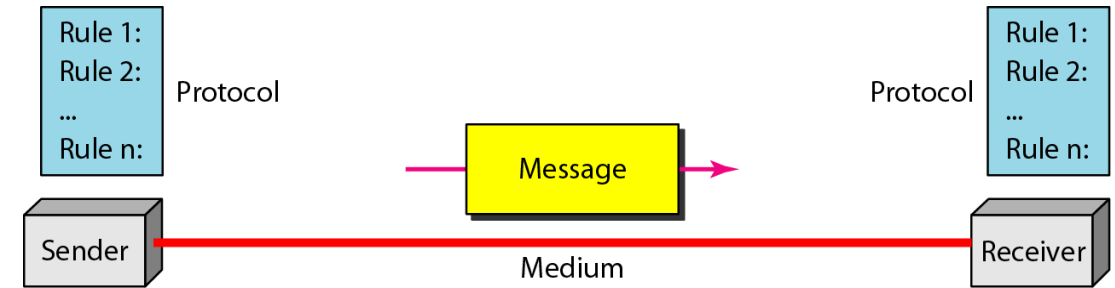
Data Communications

- *Data communications* are the exchange (sharing) of data between two devices via some form of transmission medium.
- The word *data* refers to information presented in whatever form is agreed upon by the parties creating and using the data
- The term *telecommunications* means communication over distance



Components of a Communication System

- **Sender:** Device that sends the message
- **Receiver:** Device that receives the message
- **Transmission medium:** A channel through which the message is transmitted
- **Message:** Information (data) to be communicated
- **Protocol:** The set of rules that allows communication to happen

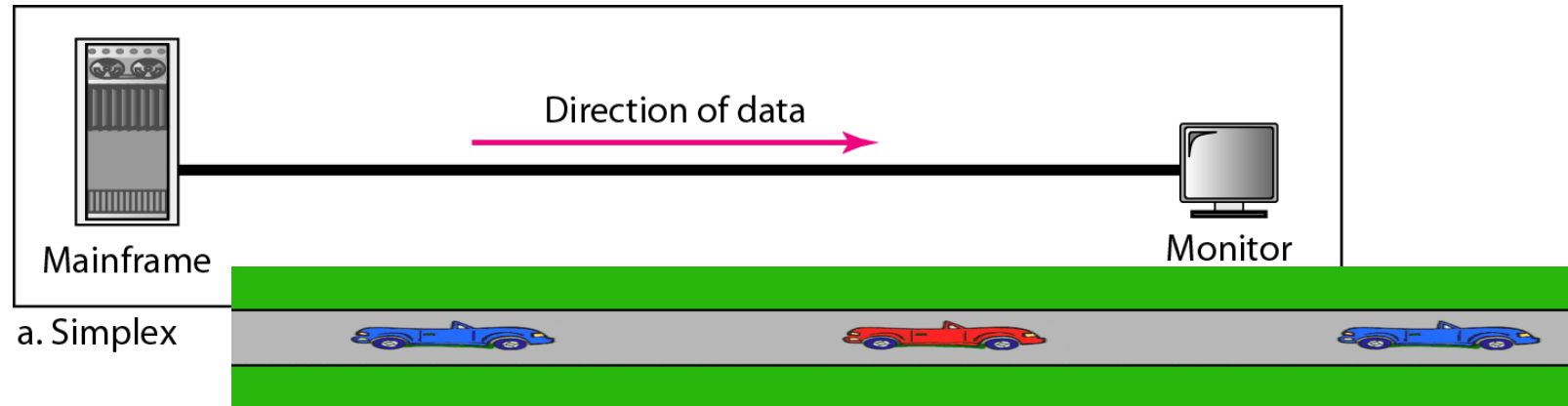


Data Flow

- *Data Flow*: The route that data takes from sender to receiver

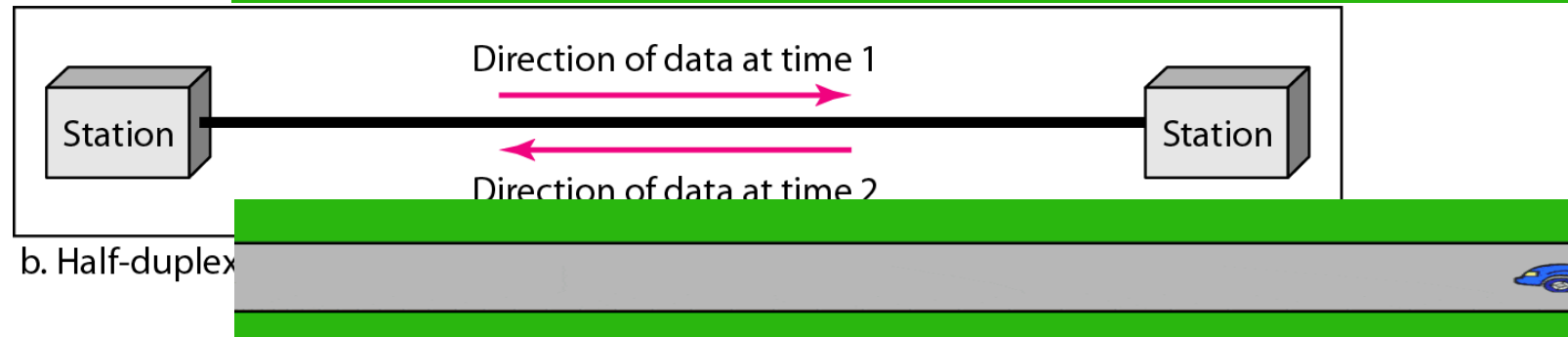
- *Simplex*:

- Keyboard
- Traditional Monitor



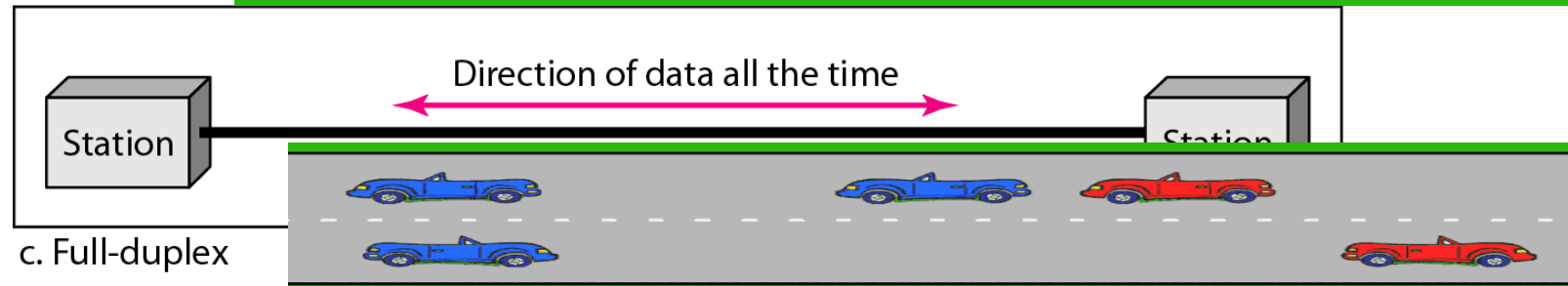
- *Half-duplex*:

- Walkie-Talkie



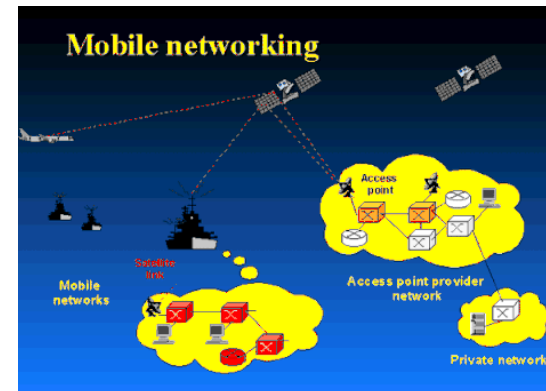
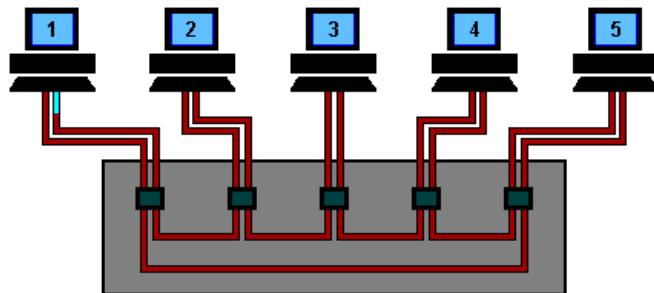
- *Full-duplex*:

- Workstations
- Upload/Download



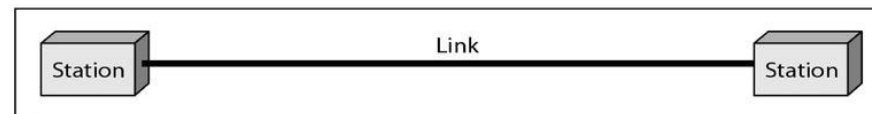
Networks

- A *network* is the interconnection of a set of devices (*nodes*) capable of communicating via communication *links*
- A *node* can be a computer, printer, or any other device
 - It is capable of sending/receiving data generated by other nodes on the network
- A *link* can be a cable, air, optical fiber, or any medium which can transport a signal carrying information.

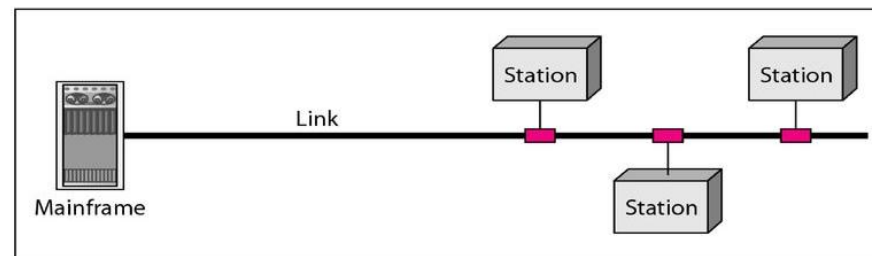


Type of connection

- A *point-to-point* provides a dedicated link between two devices. The entire capacity of the link is reserved for the transmission between these two devices.
- A *multipoint (multidrop)* connection is one in which one or more than two specific devices share a single link



a. Point-to-point



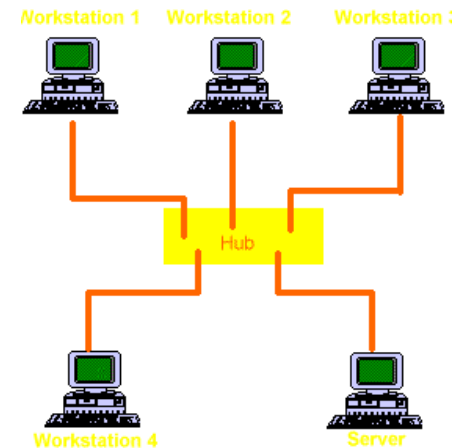
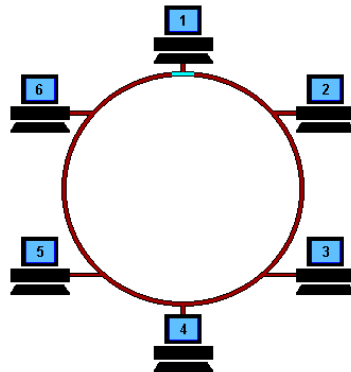
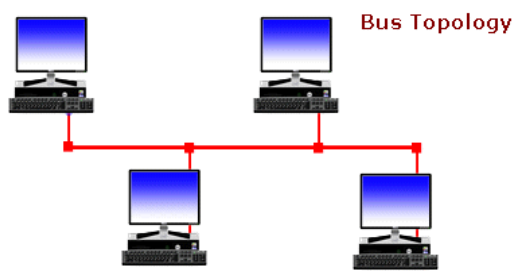
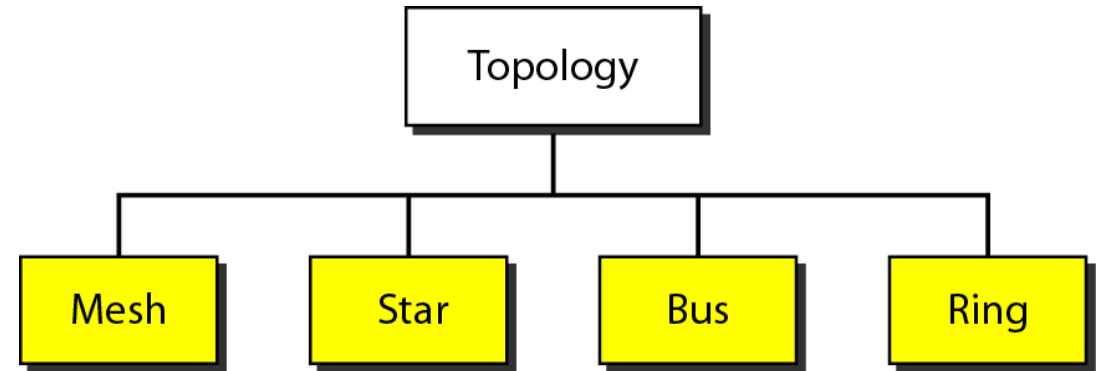
b. Multipoint

Network Topology

- **Topology:** *The **physical** layout of a network*

- **Types**

- Bus
- Ring
- Star
- Mesh
- Hybrid

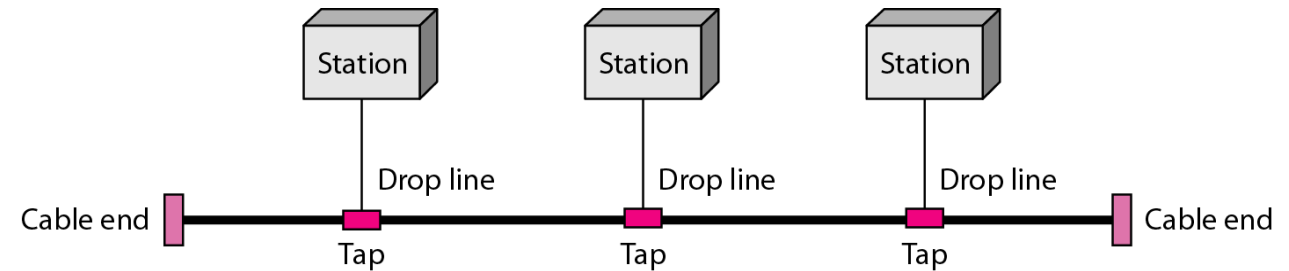


[Further details about Network Topologies](#)

Bus Topology

- *Three stations Bus*

- All data are transmitted over the common transmission medium
- All data are able to be received by all nodes in the network virtually simultaneously (disregarding propagation delay)
- The medium has exactly two endpoints called **backbone** or **trunk**

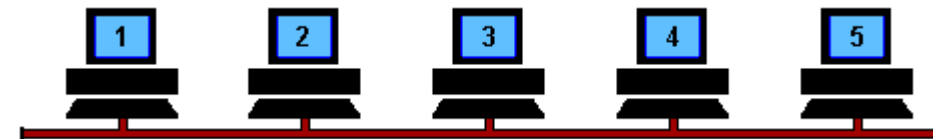


- *Five stations Bus with/without collision*

- Connected to a single cable



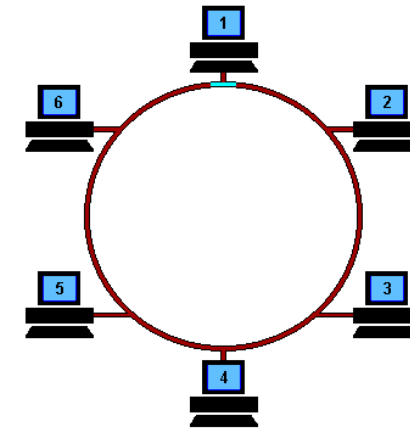
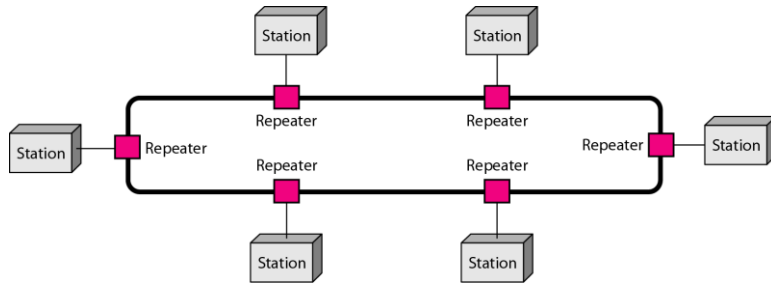
Bus topology



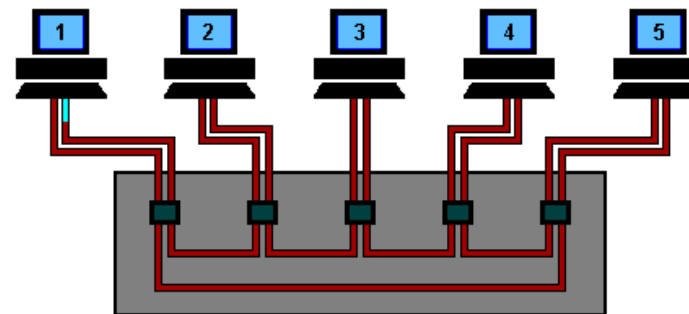
Collision in Bus

Ring Topology

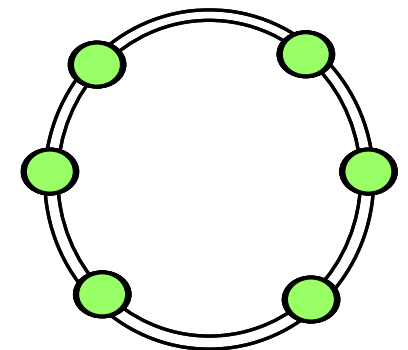
- *A ring topology connecting six stations*
- Every node is connected to the two nearest nodes so that the entire network forms a circle



- Simple and Dual Ring
 - The dual ring has twice the capacity as data can travel in both directions simultaneously



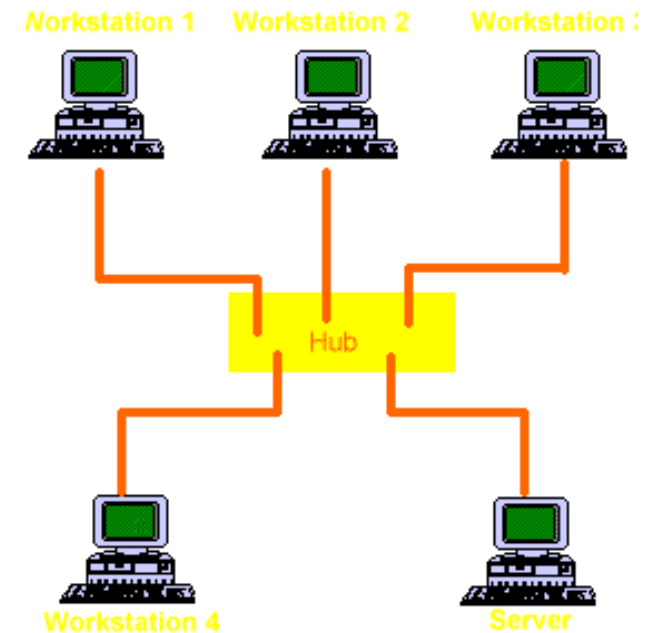
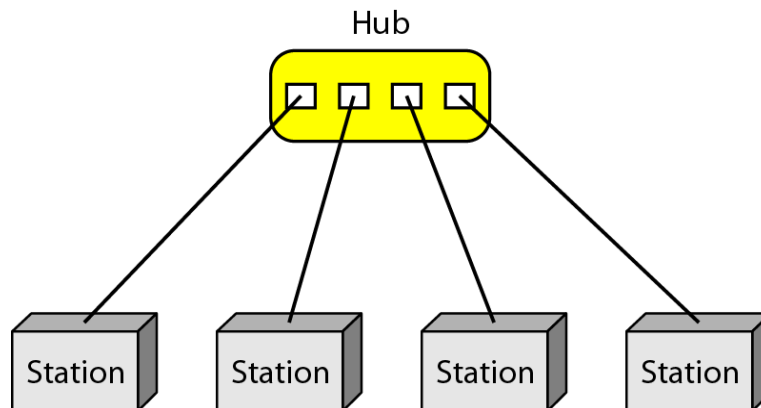
A token in the ring topology



Star Topology

- *Four stations star*

- All nodes are connected to a centralized device (*hub/switch/router*)
- The devices are not directly linked to one another
- No traffic allowed between devices



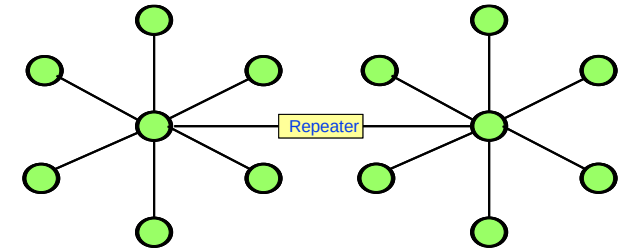
Star Topology-Variants

- *Simple Star*

- Already discussed

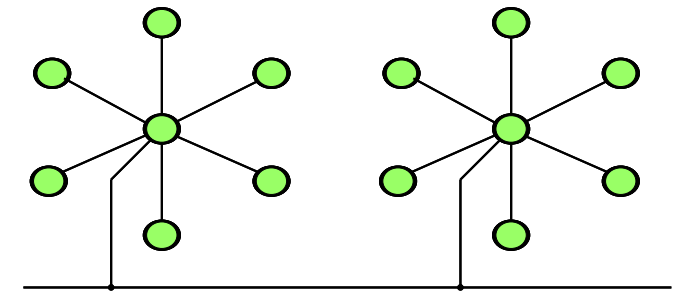
- *Extended Star*

- Has one or more repeaters between the central node
- Repeater is used to extend the maximum transmission of the simple star network



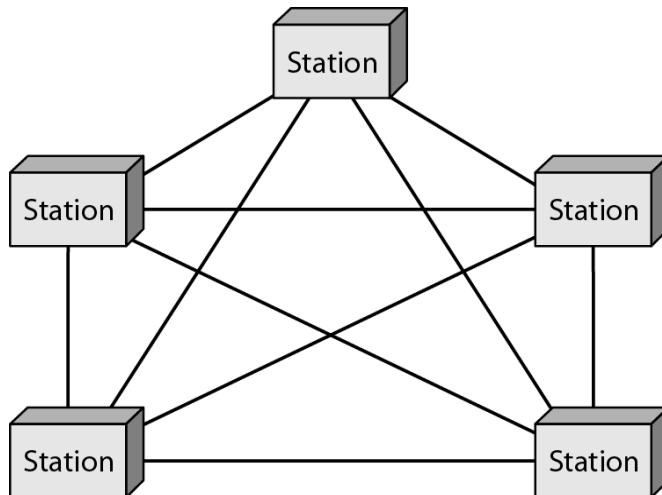
- *Distributed Star*

- Connects multiple star networks with a daisy chain in a linear fashion
- Has no hierarchy and no central connection



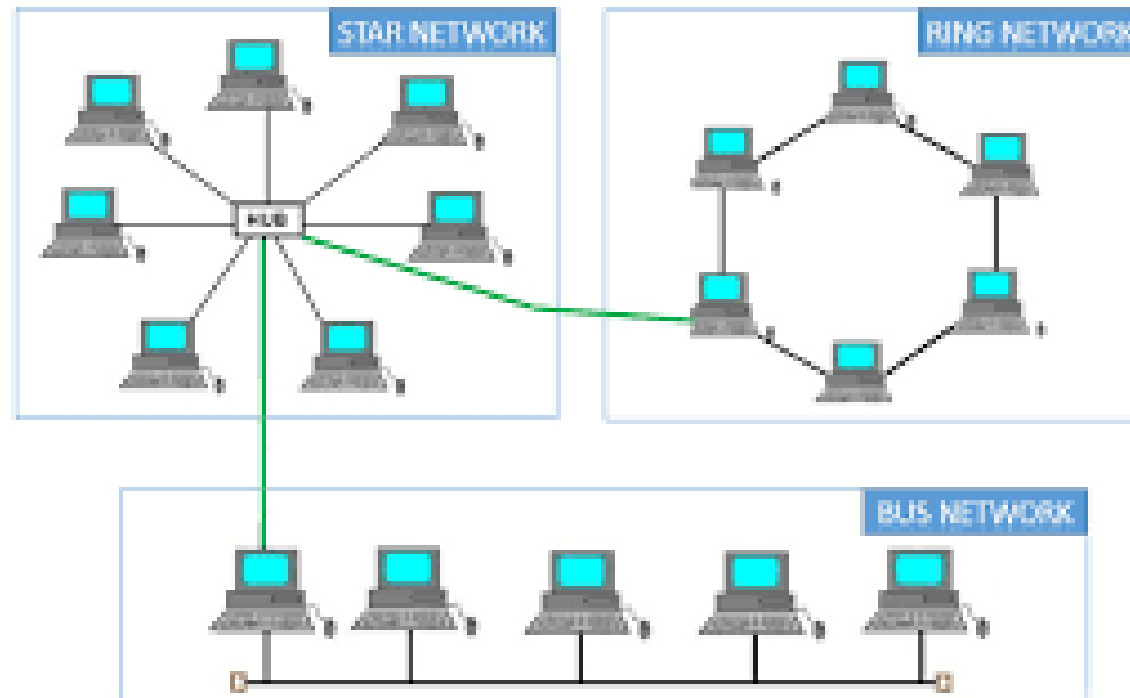
Mesh Topology

- *Five nodes Mesh:*
 - A fully connected Mesh topology with 5 nodes
- *Number of P-to-P connections required are $n(n-1)/2$ (duplex)*



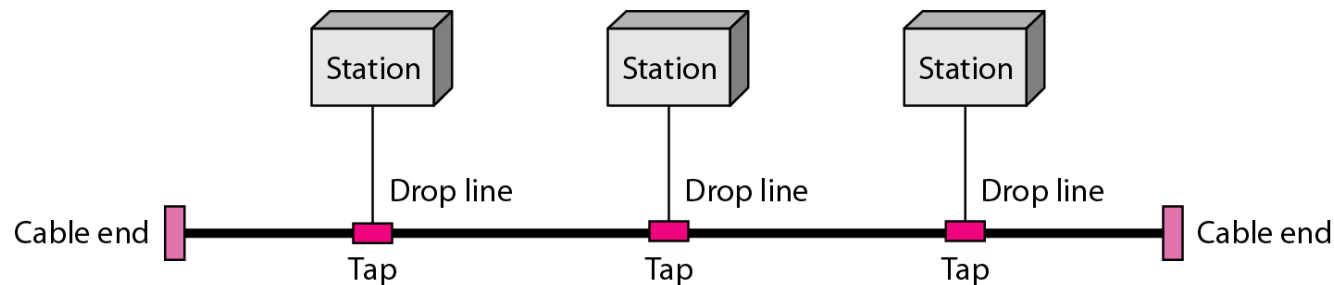
Hybrid Topology

- *Hybrid Topology:*
 - Combination of different topologies (e.g., star, ring, and bus)



Bus Topology - Pros & Cons

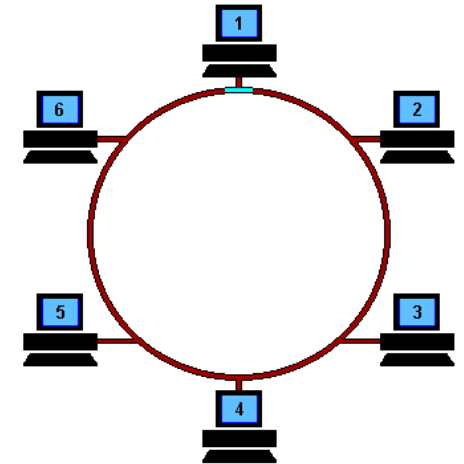
Advantages	Disadvantages
Cheap and easy to implement.	Network disruption when computers are added or removed.
Require less cable.	A break in the cable will prevent all systems from accessing the network.
Does not use any specialized network to troubleshoot.	Difficult to troubleshoot.





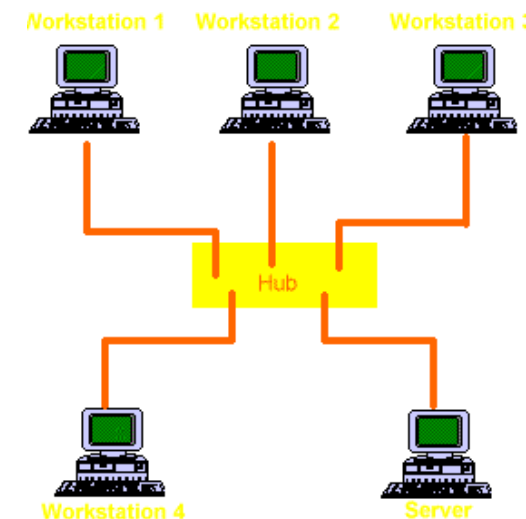
Ring Topology - Pros & Cons

Advantages	Disadvantages
Cable faults are easily located, making troubleshooting easier.	Expansion to the network can cause network disruption.
Ring networks are moderately easy to Install.	A single break in the cable can disrupt the entire network.



Star Topology – Pros & Cons

Advantages	Disadvantages
Easily expanded without disruption.	Requires more cable.
Cable failure affects only a single User.	A break in the cable of central node will prevent all systems from accessing the network.
Easy to troubleshoot and isolate problems.	More difficult to implement.



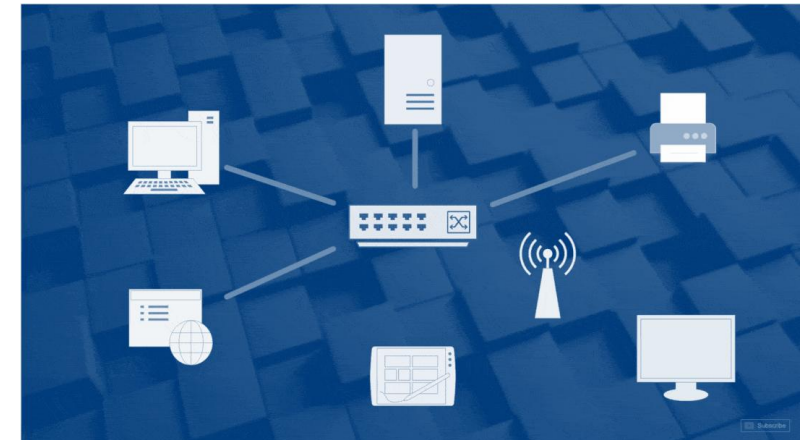
Mesh Topology – Pros & Cons

Advantages	Disadvantages
Provides redundant paths between devices.	Requires more cable than the other LAN topologies.
The network can be expanded without disruption to current users.	Complicated implementation.



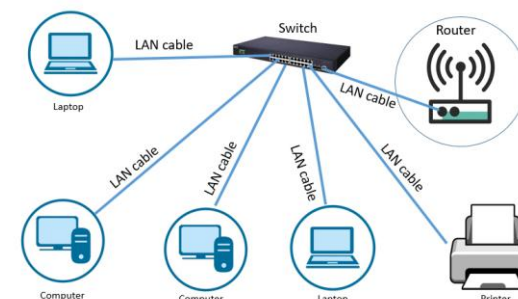
Networks Types and Categories

- Networks are categorized based on different criteria
- *Geographical span:*
 - Local Area Networks (LANs)
 - Metropolitan Area Networks (MANs)
 - Wide Area Networks (WANs)
- *Structure:*
 - Client-server
 - Peer-to-peer
- *Type of connections:*
 - Wired
 - Wireless

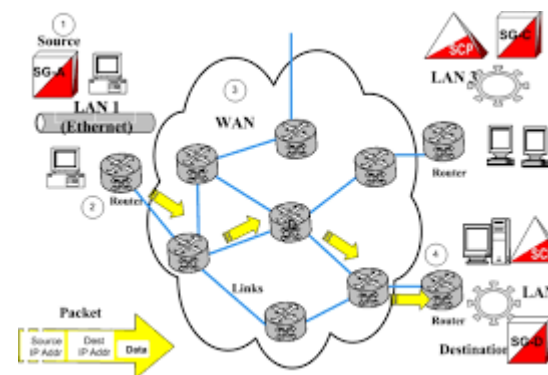
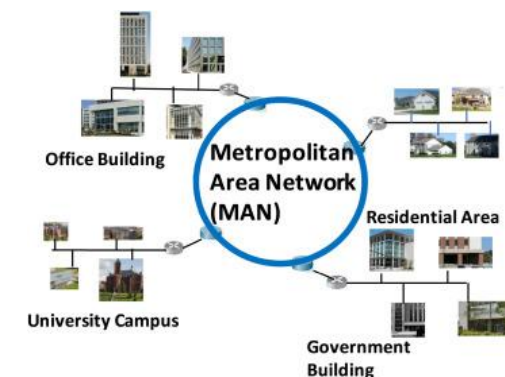


Geographical Networks

- *Local Area Networks (LANs)*
 - Short distances
 - Designed to provide local interconnectivity
- *Metropolitan Area Networks (MANs)*
 - Provide connectivity over multiple buildings
- *Wide Area Networks (WANs)*
 - Long distances
 - Provide connectivity over large area
- *Other Categories*
 - *Personal Area Network (PAN)*
 - *Storage/System Area Networks (SAN)*
 - *Body Area Network (BAN)*
 - *Campus Area Network (CAN), etc.*



Local Area Network





Geographical Networks-Classifications

Interprocessor distance	Processors located in same	Example
1 m	Square meter	Personal area network
10 m	Room	Local area network
100 m	Building	
1 km	Campus	
10 km	City	Metropolitan area network
100 km	Country	Wide area network
1000 km	Continent	
10,000 km	Planet	The Internet

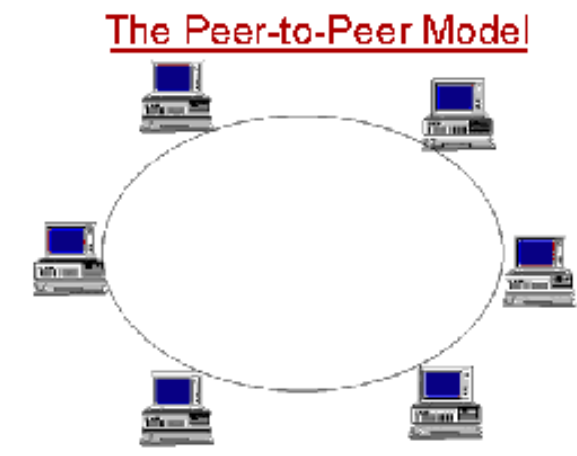
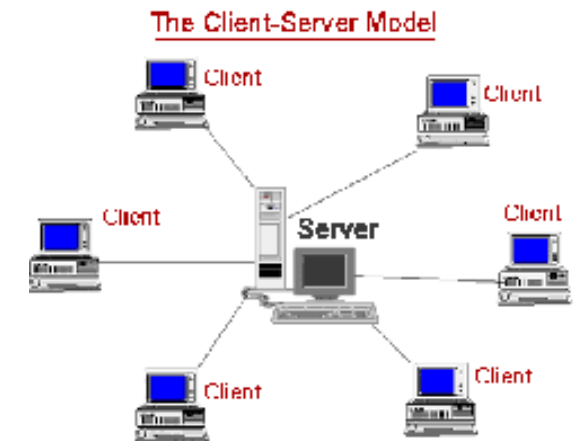
Network Structure

- *Client-server:*

- Acts as a medium to access services and resources from a central computer using a WAN or LAN
- The Client-server network is mainly used for game hosting, web services, and personal networks

- *Peer-to-peer:*

- Authorizes to link computers to a single system
- A distributed application architecture that splits tasks between peers
- A peer is a node that delivers the exact functionality as another
 - For example, two PCs in a network are peers.





Network Connections-Wired

- *Ethernet Networks:*

- The most common type for creating LANs
- They employ Ethernet cables, which are usually made up of copper, to connect devices like computers, routers, and switches

- *Fiber Optic Networks:*

- They use light signals to transmit data over thin strands of glass or plastic known as fiber optic cables
- They are known for their high-speed data transfer capability

- *Coaxial Cable Networks:*

- These were used in cable television and early computer networks



Network Connections-Wireless

- *Wi-Fi Networks:*
 - The most well-known type of wireless networks
 - It uses radio waves to connect devices to the internet or each other
- *Bluetooth Networks:*
 - They enable communication between devices over short distances
 - They're typically used for connecting peripherals like wireless mice, keyboards, or headphones to a computer or smartphone
- *Cellular Networks:*
 - These are used by mobile phones and some IoT devices
 - They have a large network of fixed-location cell towers
 - These networks support voice calls, SMS, and data services, such as 4G and 5G internet access



The Internet

- The network of networks is called the **Internet**
- It has revolutionized many aspects of our daily lives.
 - Can you imagine living without internet connection? How would your life be?
- The **Internet** is a data communication system that has brought a wealth of information to our fingertips and organized it for our use.

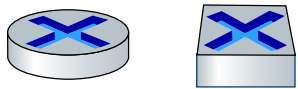


The Internet: a “nuts and bolts” view



Billions of connected computing *devices*:

- *hosts* = end systems
- running *network apps* at Internet’s “edge”



Packet switches: forward packets (chunks of data)

- *routers, switches*

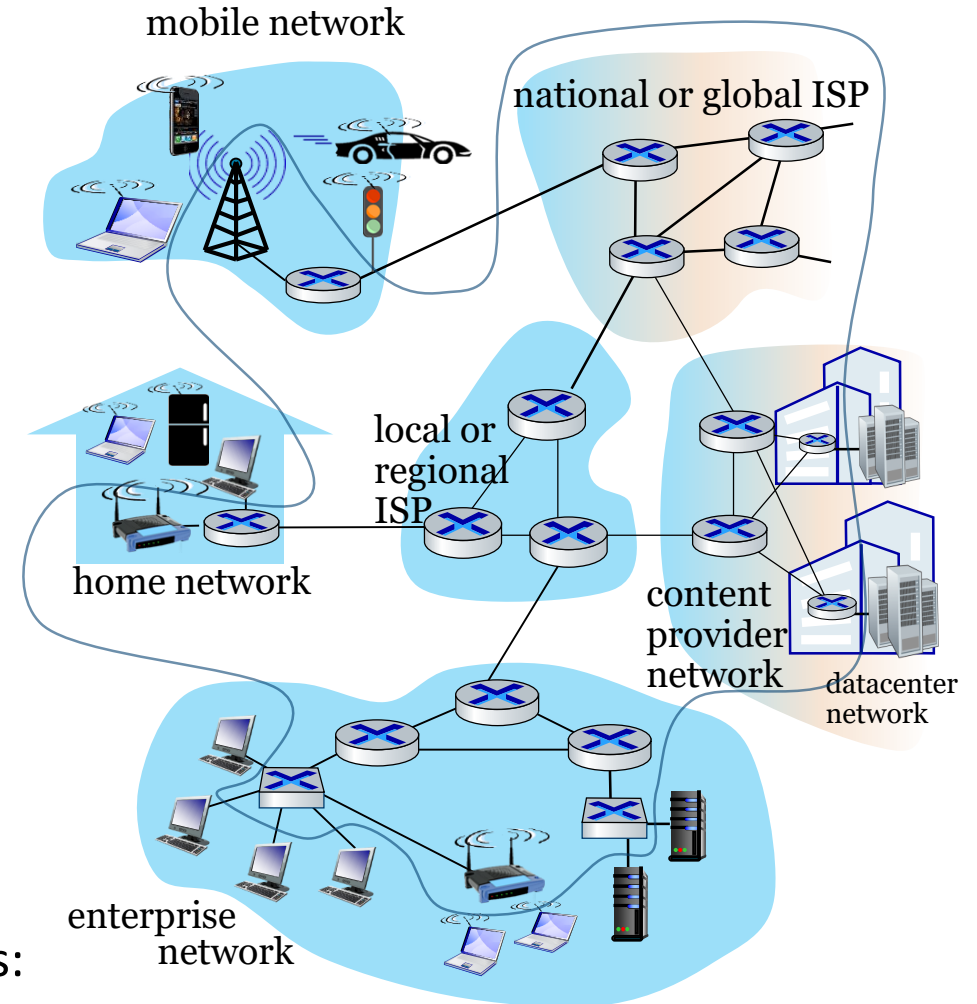
Communication links

- fiber, copper, radio, satellite
- transmission rate: *bandwidth*



Networks

- collection of devices, routers, links: managed by an organization





“Fun” Internet-connected devices



Amazon Echo



Internet
refrigerator



IP picture frame



Pacemaker & Monitor



Tweet-a-watt:
monitor energy use

bikes



Security Camera



Slingbox: remote
control cable TV



Web-enabled toaster +
weather forecaster



cars



scooters



Internet phones



Gaming devices



sensorized
bed
mattress



AR devices



Fitbit

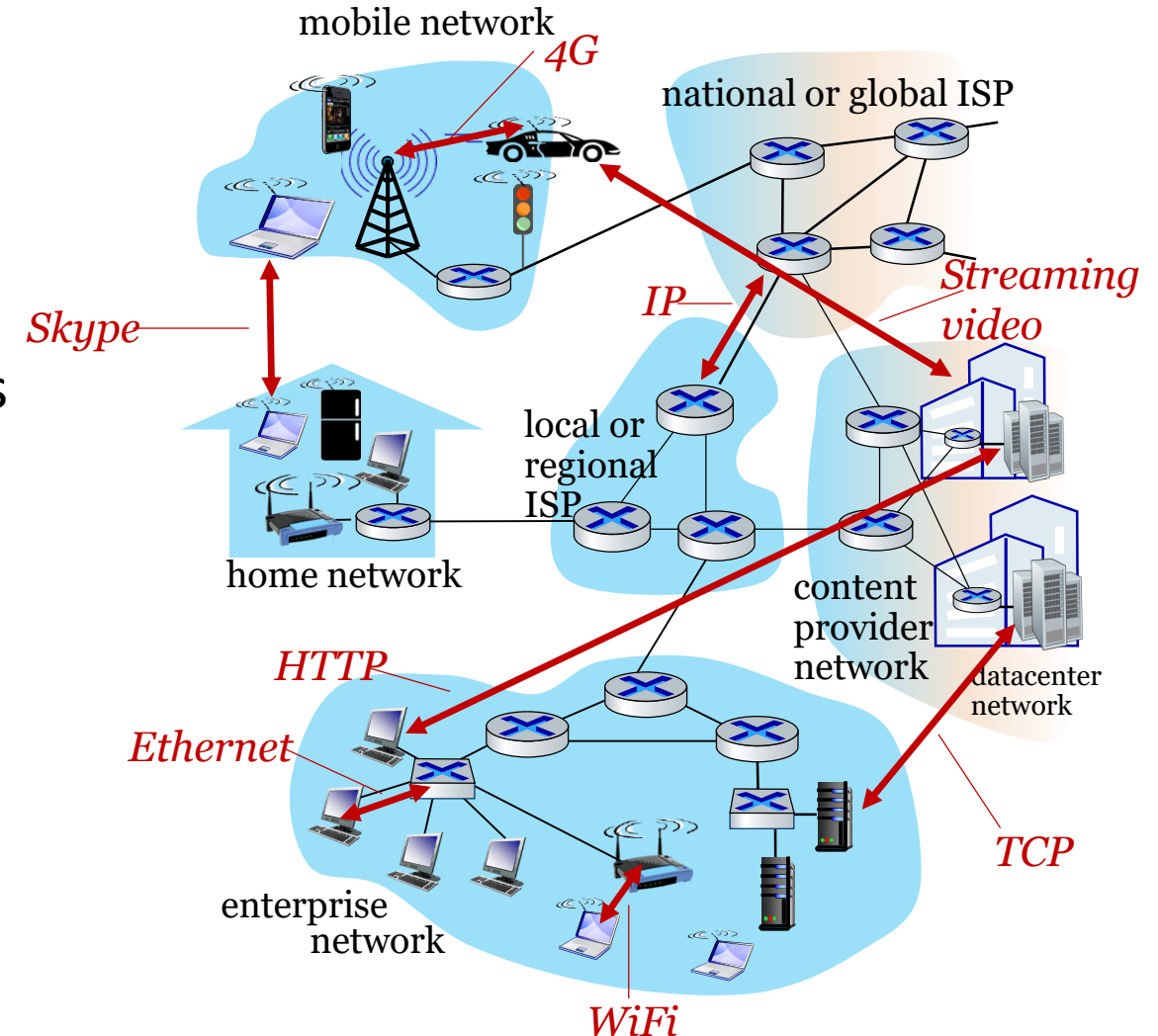


diapers

Others?

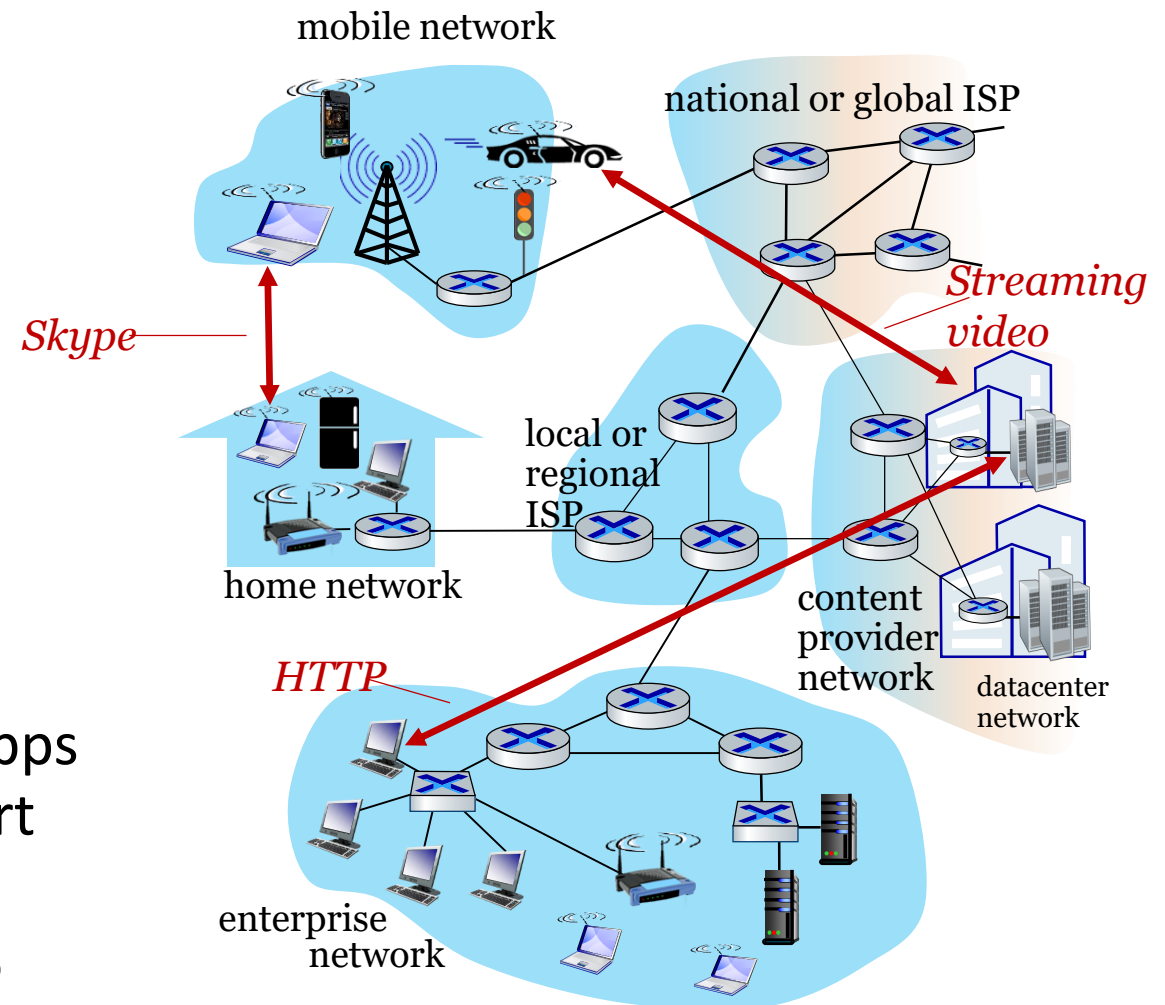
The Internet: a “nuts and bolts” view

- *Internet: “network of networks”*
 - Interconnected ISPs
 - Internet Service Providers
- *Protocols* are everywhere
 - control sending, receiving of messages
 - e.g., HTTP (Web), streaming video, Skype, TCP, IP, WiFi, 4/5G, Ethernet
- *Internet standards*
 - RFC: Request for Comments
 - <https://www.rfc-editor.org>
 - IETF: Internet Engineering Task Force



The Internet: a “services” view

- *Infrastructure* that provides services to applications:
 - Web, streaming video, multimedia teleconferencing, email, games, e-commerce, social media, inter-connected appliances, ...
- Provides *programming interface* to distributed applications:
 - “hooks” allowing sending/receiving apps to “connect” to, use Internet transport service
 - provides service options, analogous to postal service





What's a protocol?

Human protocols:

- “what’s the time?”
- “I have a question”
- introductions

Rules for:

- ... specific messages sent
- ... specific actions taken
when message received,
or other events

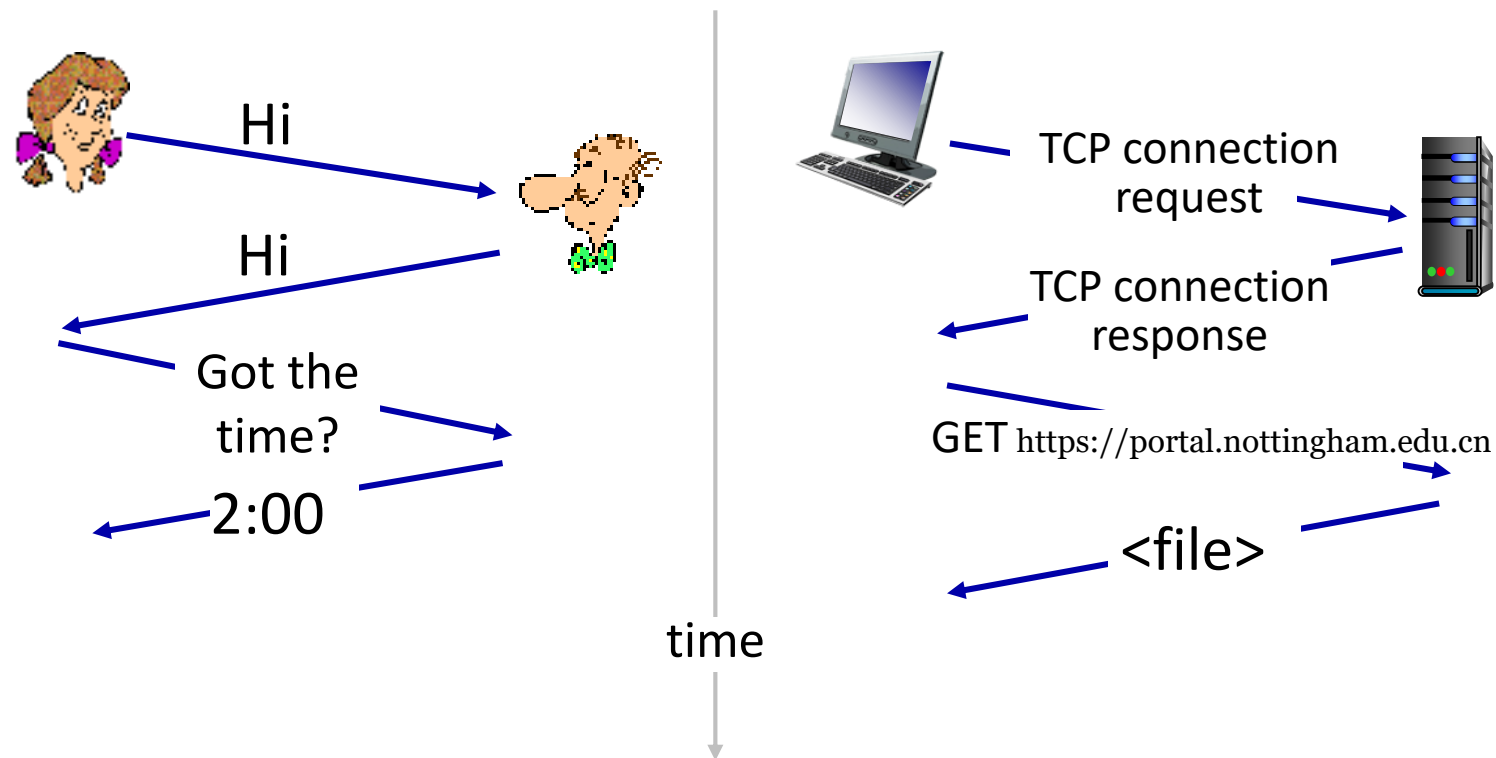
Network protocols:

- computers (devices) rather than humans
- all communication activity in Internet
governed by protocols

*Protocols define the **format, order** of
messages sent and received among
network entities, and **actions taken**
on message transmission, receipt*

What's a protocol?

A human protocol and a computer network protocol:

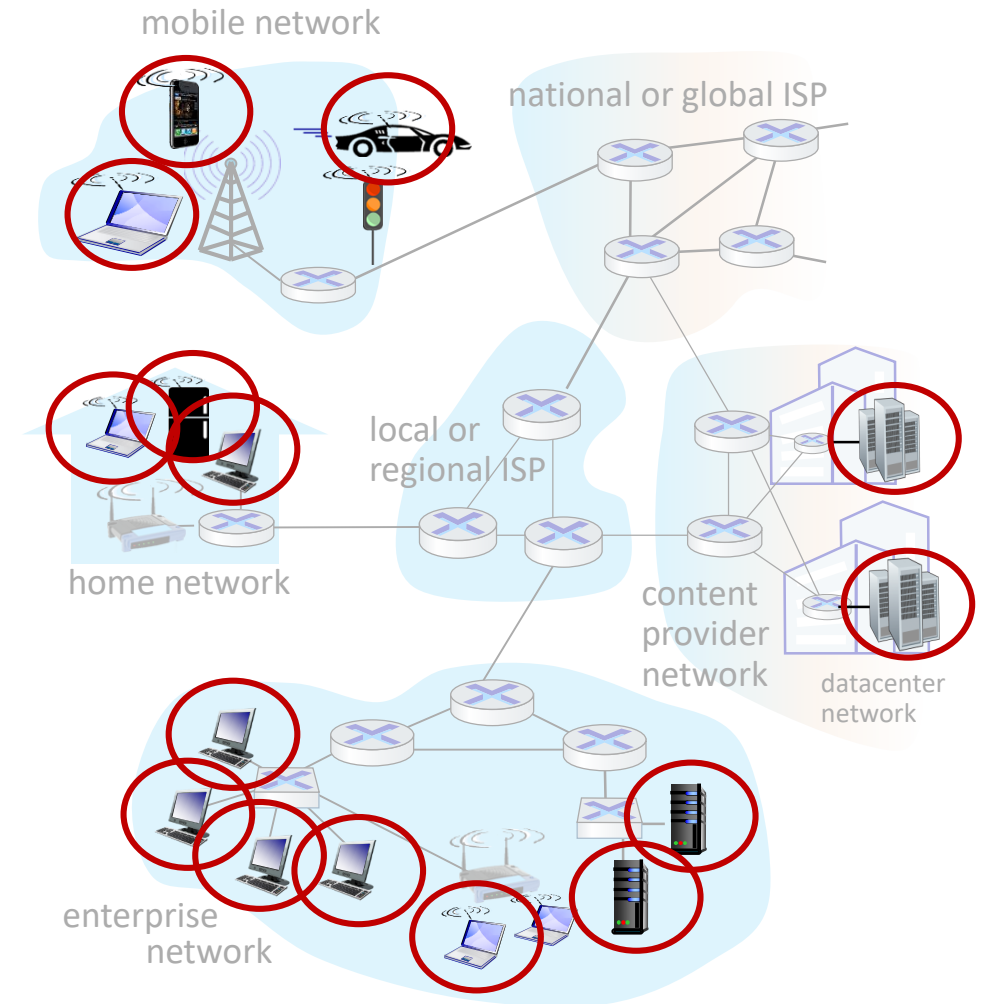


Q: other human protocols?

A closer look at Internet structure

Network edge:

- hosts: clients and servers
- servers often in data centers



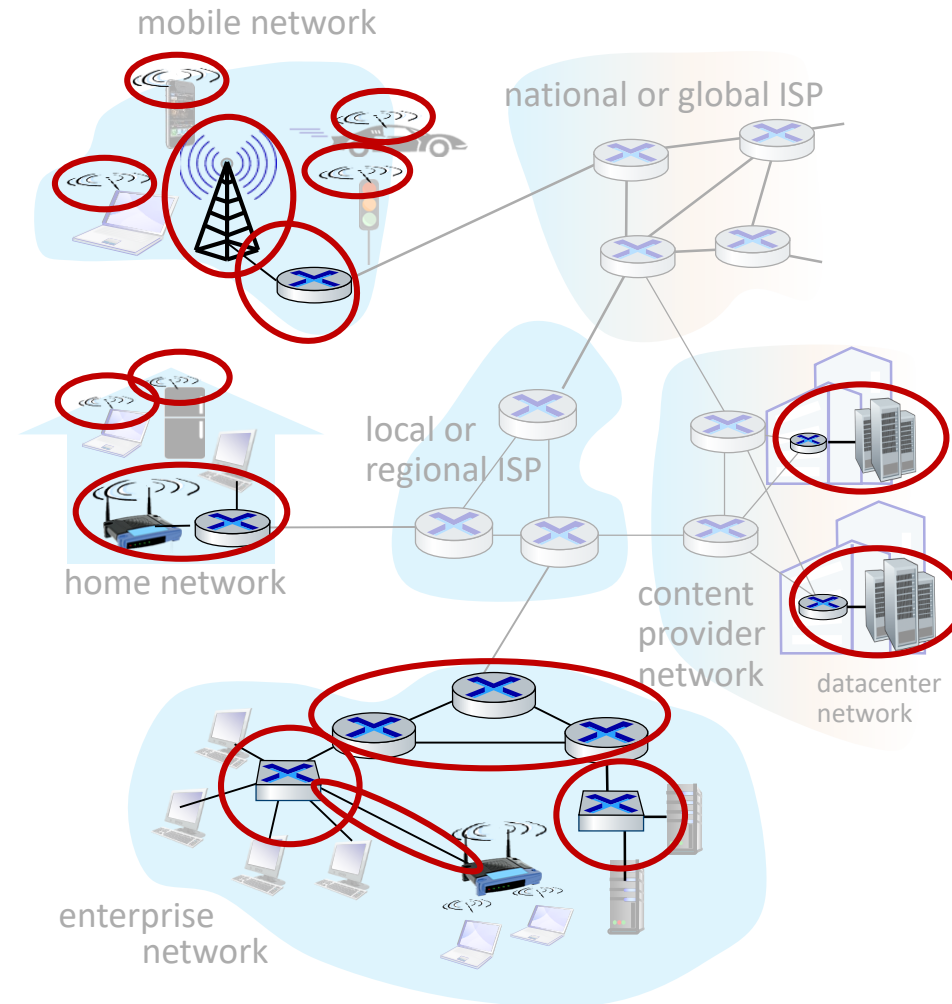
A closer look at Internet structure

Network edge:

- hosts: clients and servers
- servers often in data centers

Access networks, physical media:

- wired, wireless communication links



A closer look at Internet structure

Network edge:

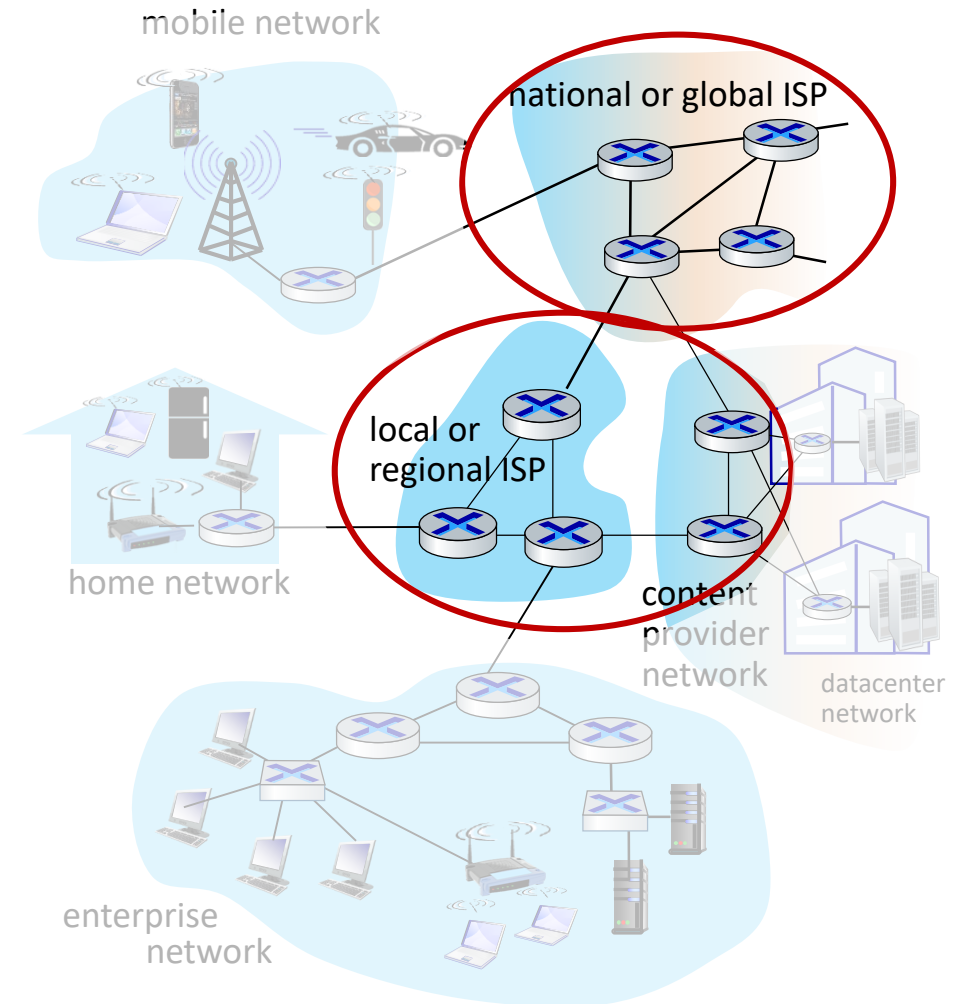
- hosts: clients and servers
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Access networks, physical media:

- wired, wireless communication links

Network core:

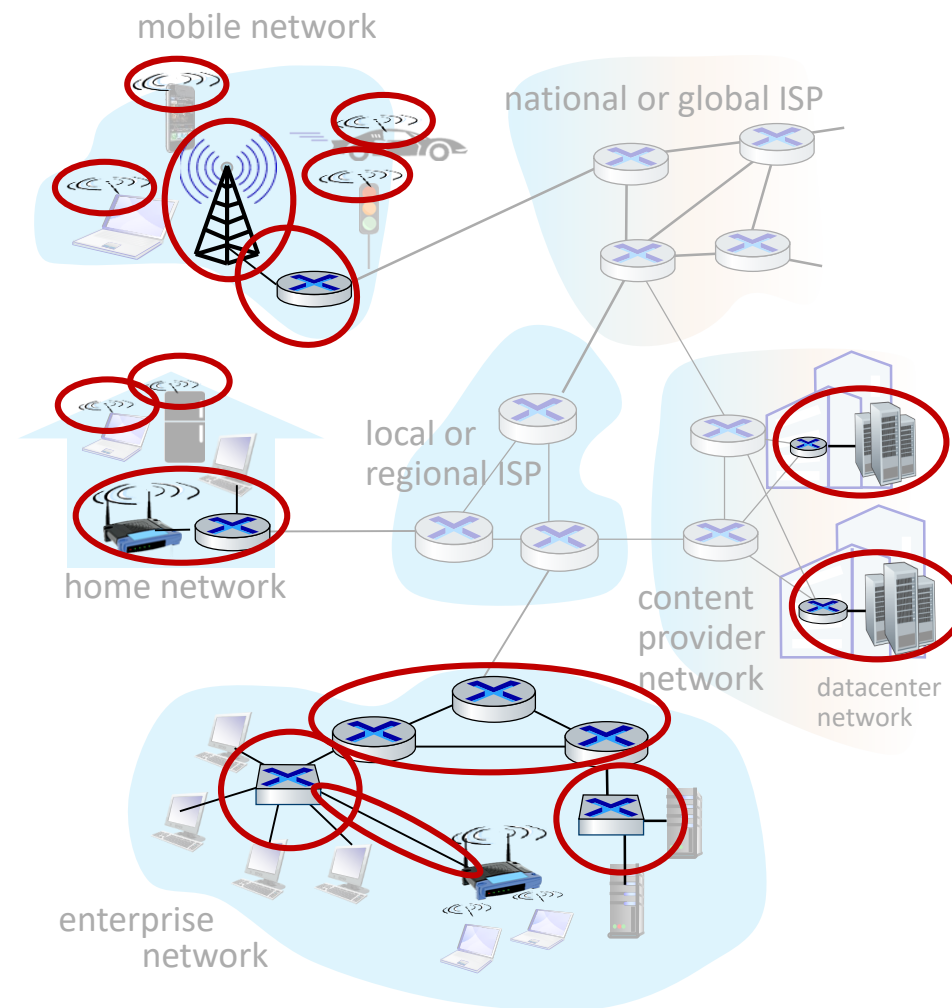
- interconnected routers
- network of networks



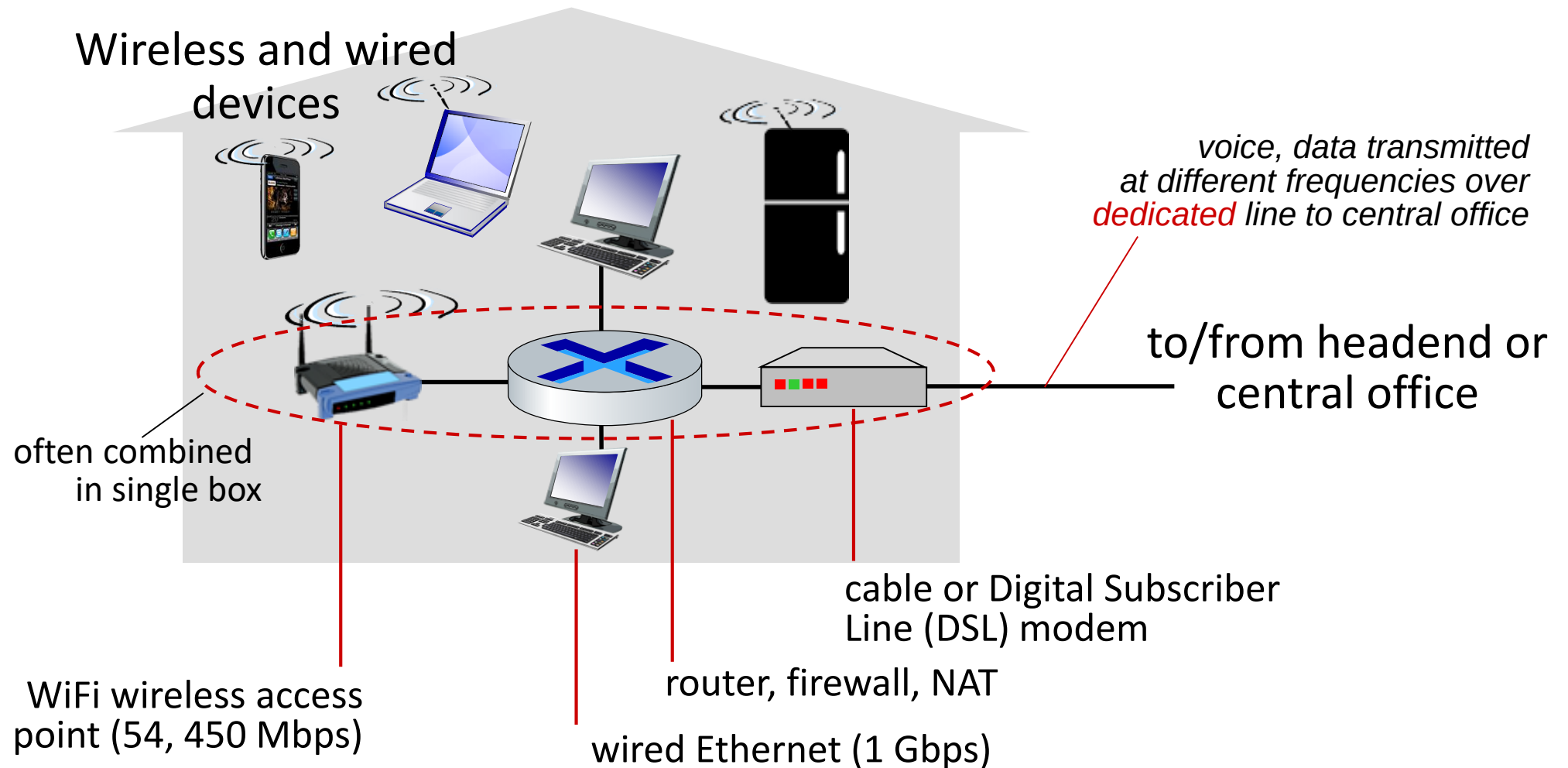
Access networks and physical media

Q: How to connect end systems to edge router?

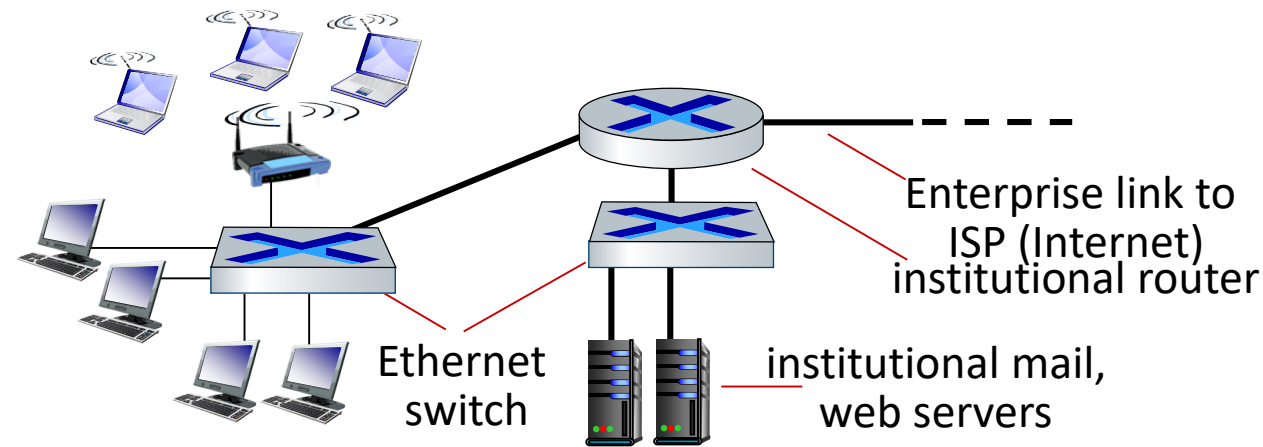
- residential access nets
- institutional access networks (school, company)
- mobile access networks (WiFi, 4G/5G)



Access networks: home networks



Access networks: enterprise networks



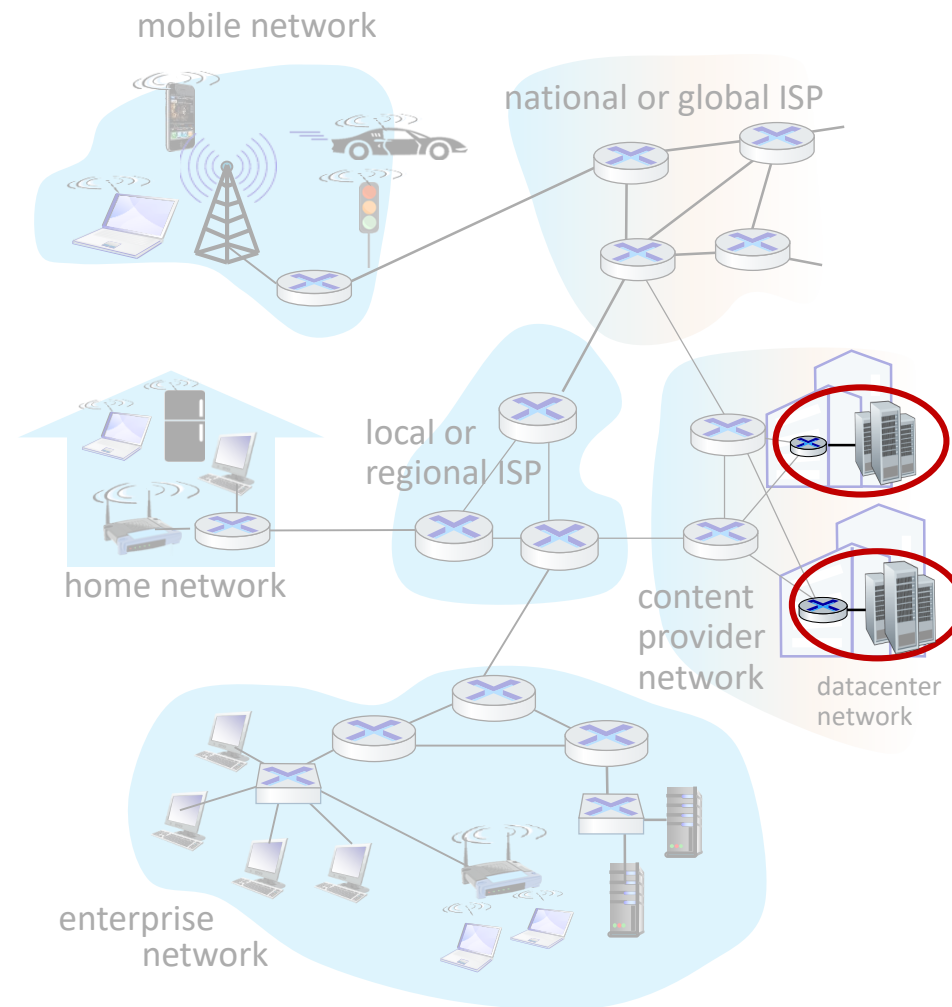
- companies, universities, etc.
- mix of wired, wireless link technologies, connecting a mix of switches and routers
- Ethernet: wired access at 100Mbps, 1Gbps, 10Gbps
- WiFi: wireless access points at 11, 54, 450 Mbps

Access networks: data center networks

- high-bandwidth links (10s to 100s Gbps) connect hundreds to thousands of servers together, and to Internet



Courtesy: Massachusetts Green High Performance Computing Center (mghpcc.org)

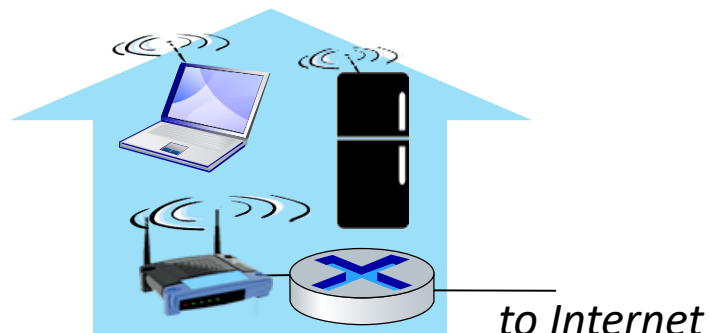


Wireless access networks

- Shared *wireless* access network connects end system to router via base station aka “access point”

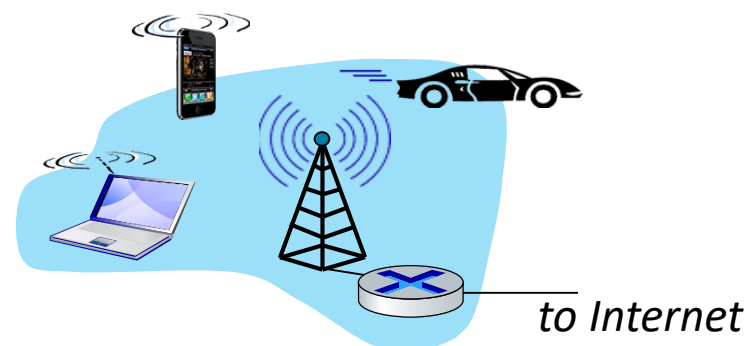
Wireless local area networks (WLANs)

- typically within or around building (~100 ft)
- 802.11b/g/n (WiFi): 11, 54, 450 Mbps transmission rate



Wide-area cellular access networks

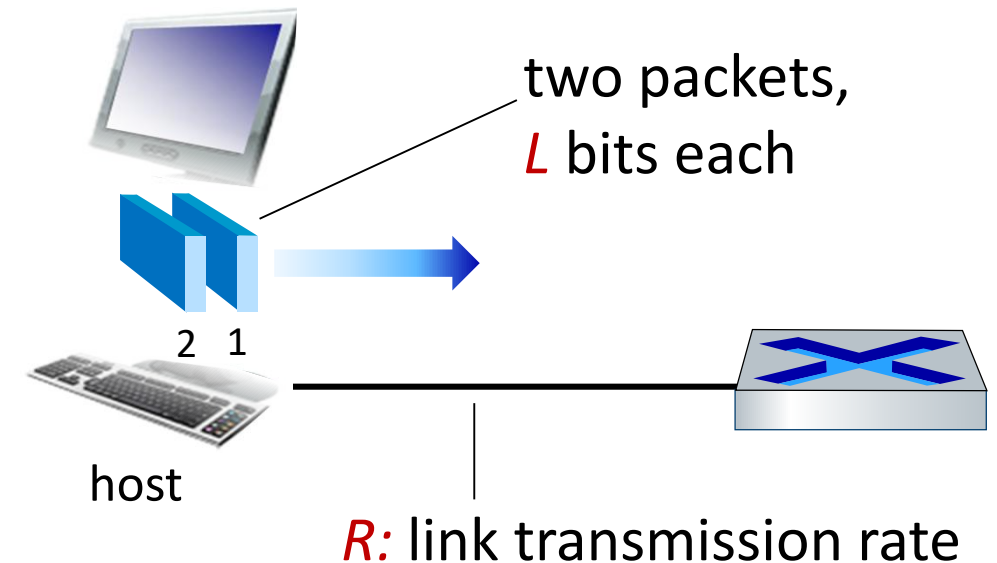
- provided by mobile, cellular network operator (10's km)
- 10's Mbps
- 4G/5G cellular networks



Host: sends packets of data

host sending function:

- takes application message
- breaks into smaller chunks, known as *packets*, of length L bits
- transmits packet into access network at *transmission rate R*
 - link transmission rate, aka link *capacity, aka link bandwidth*



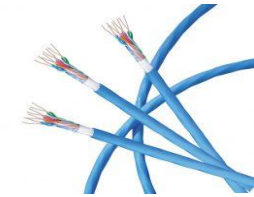
$$\begin{array}{l} \text{packet} \\ \text{transmission} \\ \text{delay} \end{array} = \begin{array}{l} \text{time needed to} \\ \text{transmit } L\text{-bit} \\ \text{packet into link} \end{array} = \frac{L \text{ (bits)}}{R \text{ (bits/sec)}}$$

Links: physical media

- **bit**: propagates between transmitter/receiver pairs
- **physical link**: what lies between transmitter & receiver
- **guided media**:
 - signals propagate in solid media: copper, fiber, coaxial
- **unguided media**:
 - signals propagate freely, e.g., radio

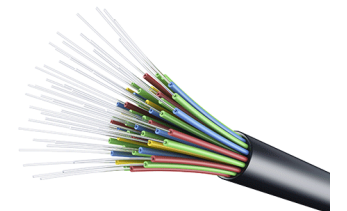
Twisted pair (TP)

- two insulated copper wires
 - Category 5: 100 Mbps, 1 Gbps Ethernet
 - Category 6: 10Gbps Ethernet



Fiber optic cable:

- glass fiber carrying light pulses, each pulse a bit
- high-speed operation:
 - high-speed point-to-point transmission (10's-100's Gbps)
- low error rate:
 - repeaters spaced far apart
 - immune to electromagnetic noise



Links: physical media

Wireless radio

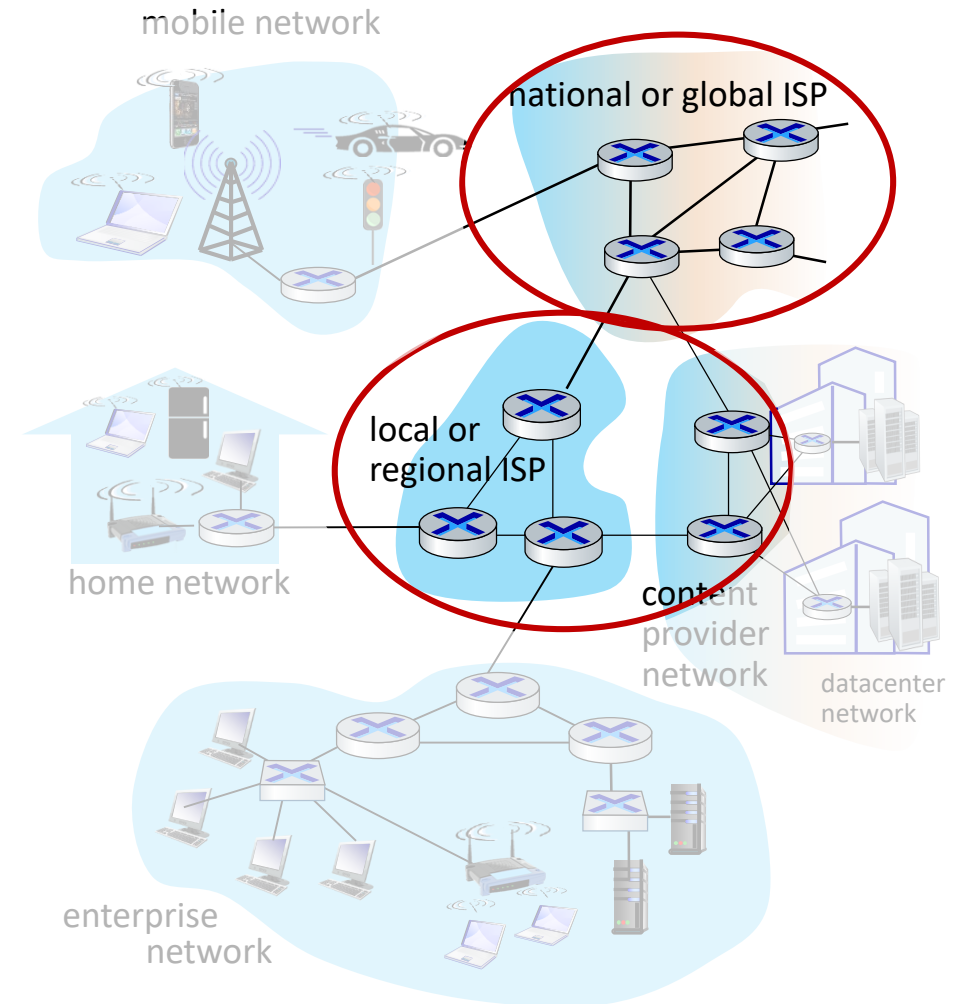
- signal carried in various “bands” in electromagnetic spectrum
- no physical “wire”
- broadcast, “half-duplex” (sender to receiver)
- propagation environment effects:
 - reflection
 - obstruction by objects
 - Interference/noise

Radio link types:

- **Wireless LAN (WiFi)**
 - 10-100’s Mbps; 10’s of meters
- **Wide-area** (e.g., 4G/5G cellular)
 - 10’s Mbps (4G) over ~10 Km
- **Bluetooth**: cable replacement
 - short distances, limited rates
- **Terrestrial microwave**
 - point-to-point; 45 Mbps channels
- **Satellite**
 - up to < 100 Mbps (Starlink) downlink
 - 270 msec end-end delay (geostationary)

The network core

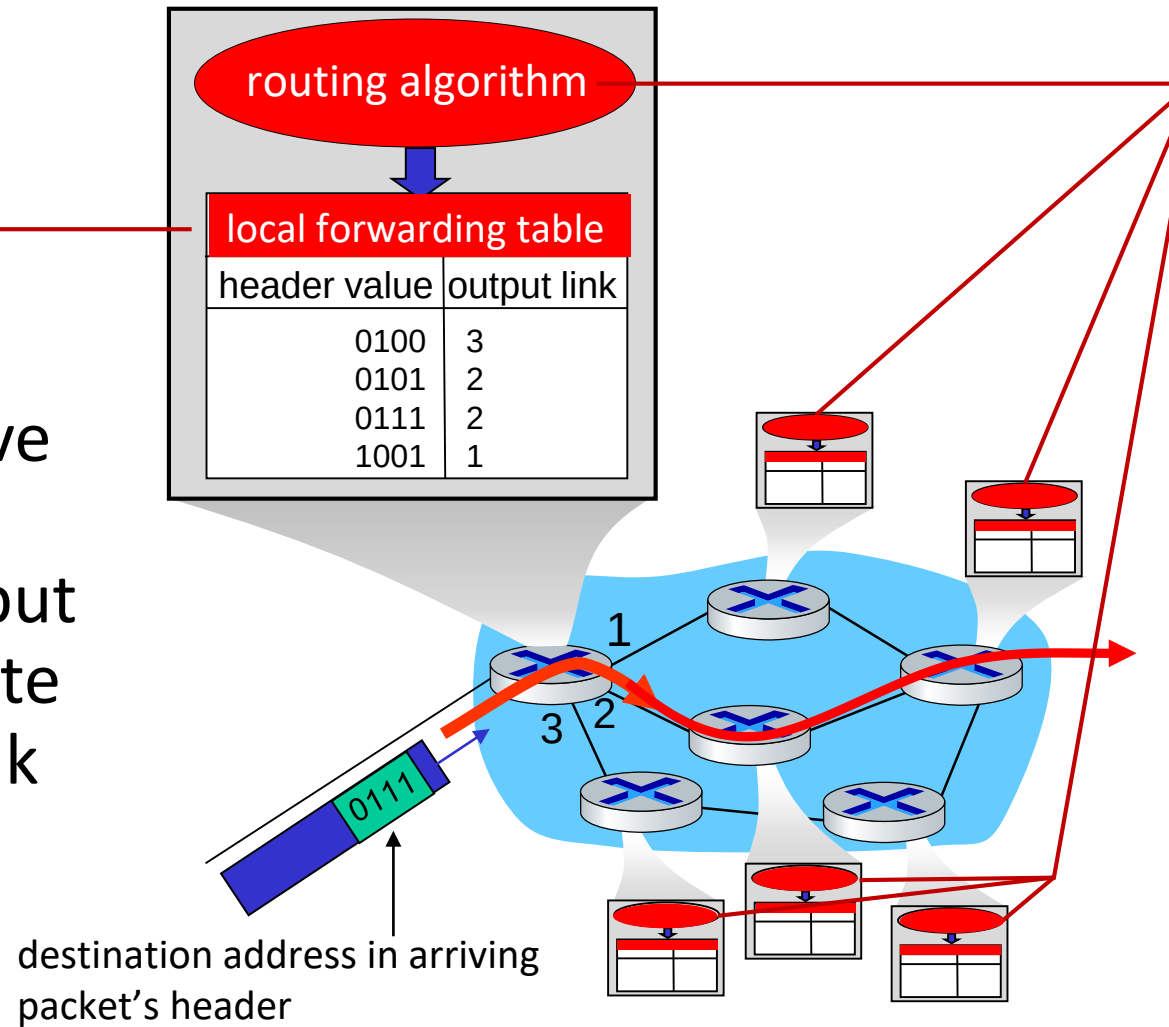
- mesh of interconnected routers
- **packet-switching**: hosts break application-layer messages into *packets*
 - network **forwards** packets from one router to the next, across links on path from **source to destination**



Two key network-core functions

Forwarding:

- aka “switching”
- *local* action: move arriving packets from router’s input link to appropriate router output link



Routing:

- *global* action: determine source-destination paths taken by packets
- routing algorithms



Routing





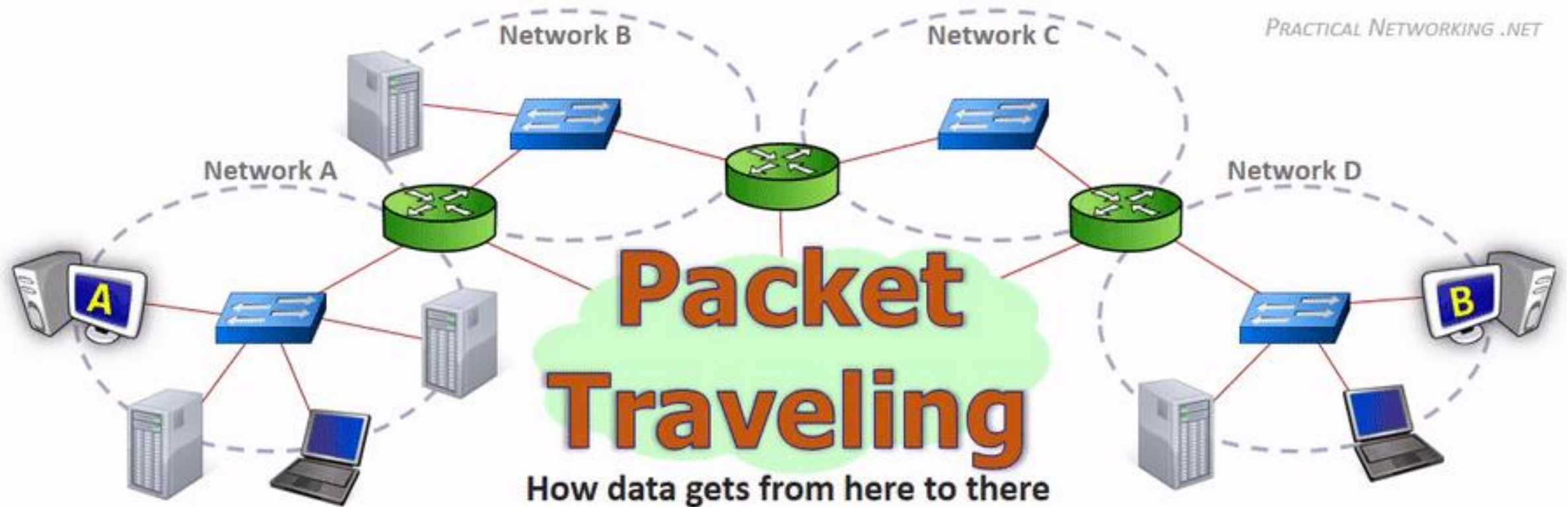
Forwarding



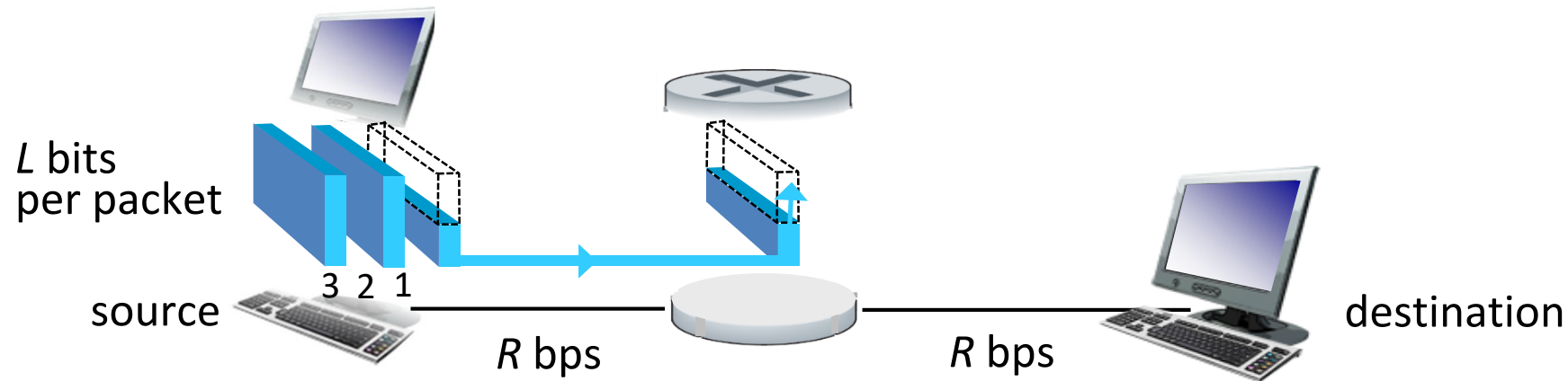
Packet Traveling

Tracert/Traceroute Format

Hop RTT1 RTT2 RTT3 Domain Name [IP Address]



Packet-switching: store-and-forward

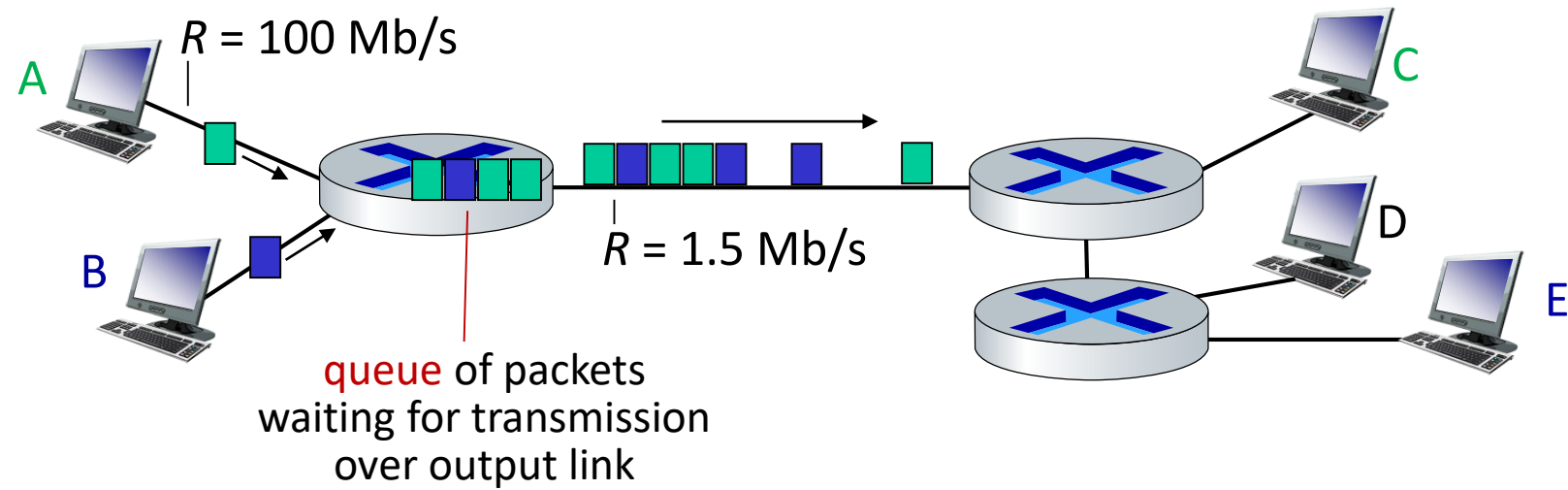


- **packet transmission delay:** takes L/R seconds to transmit (push out) L -bit packet into link at R bps
- **store and forward:** entire packet must arrive at router before it can be transmitted on next link

One-hop numerical example:

- $L = 10$ Kbits
- $R = 100$ Mbps
- one-hop transmission delay = 0.1 msec

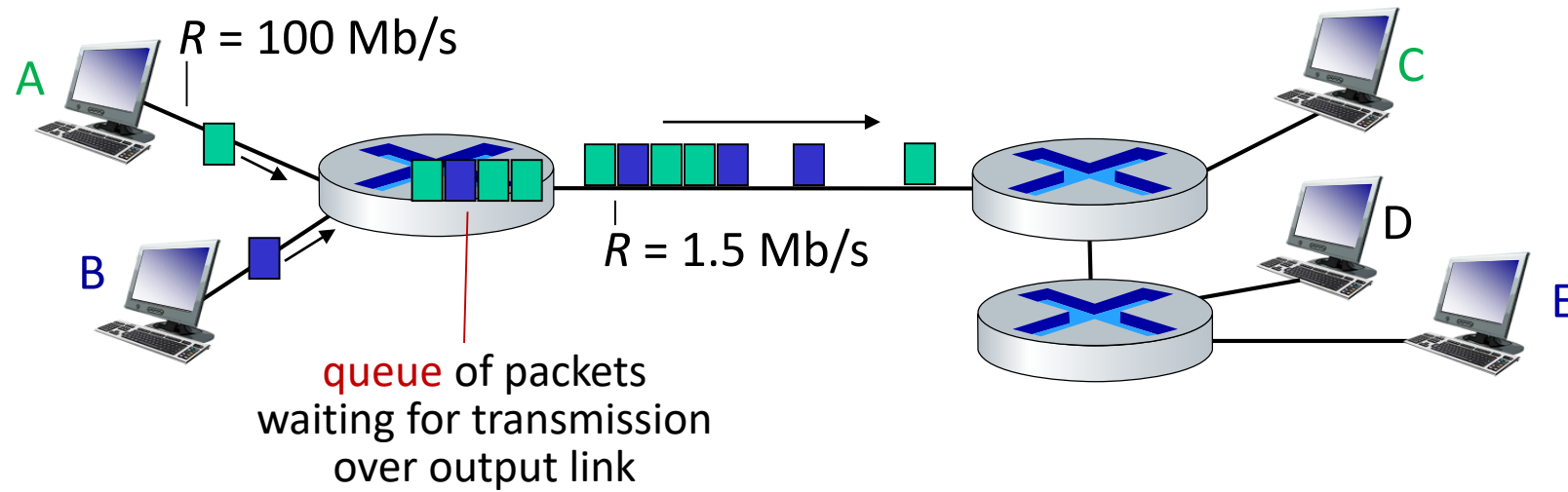
Packet-switching: queueing



Queueing occurs when work arrives faster than it can be serviced:



Packet-switching: queueing



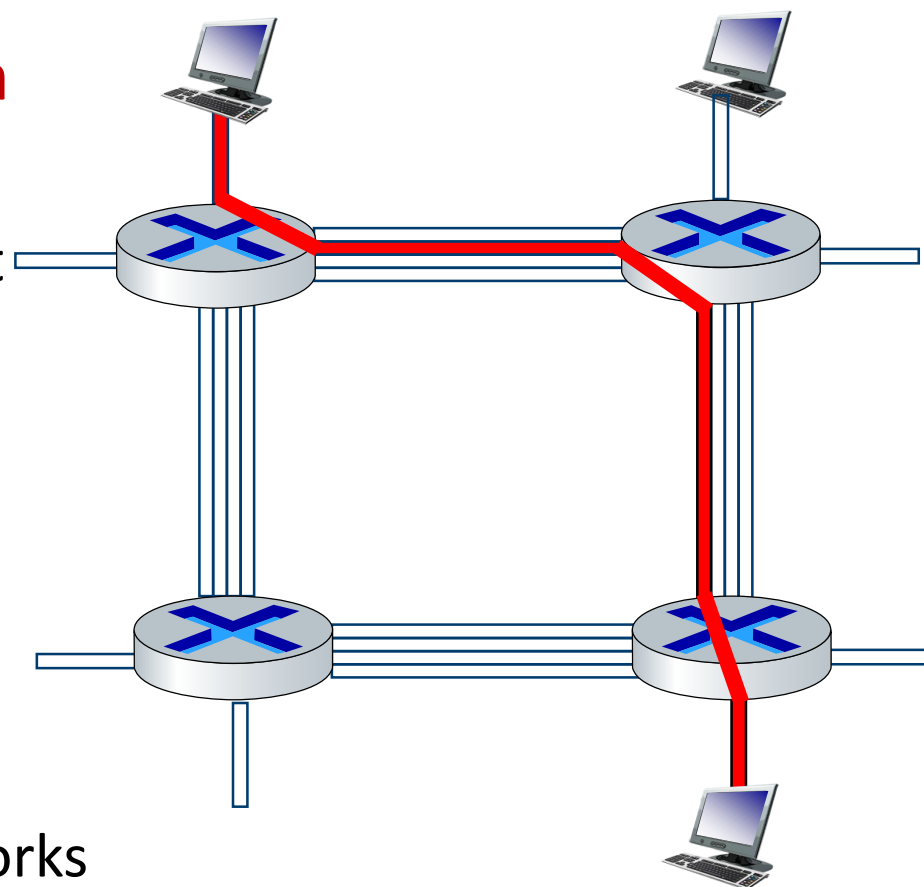
Packet queueing and loss: if arrival rate (in bps) to link exceeds transmission rate (bps) of link for some period of time:

- packets will queue, waiting to be transmitted on output link
- packets can be dropped (lost) if memory (buffer) in router fills up

Alternative to packet switching: circuit switching

End-end resources allocated to, reserved for “call” between source and destination

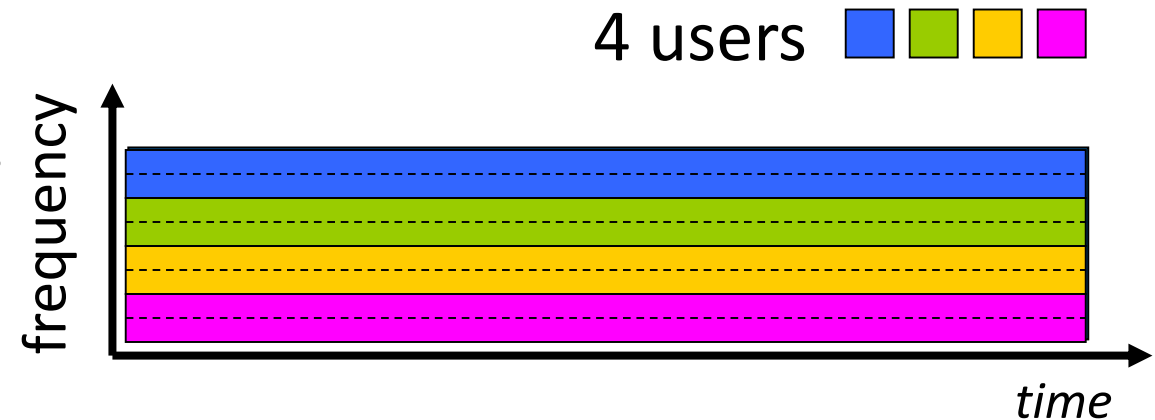
- in diagram, each link has four circuits.
 - call gets 2nd circuit in top link and 1st circuit in right link.
 - dedicated resources: no sharing
 - circuit-like (guaranteed) performance
 - circuit segment idle if not used by call (no sharing)
- commonly used in traditional telephone networks



Circuit switching: FDM and TDM

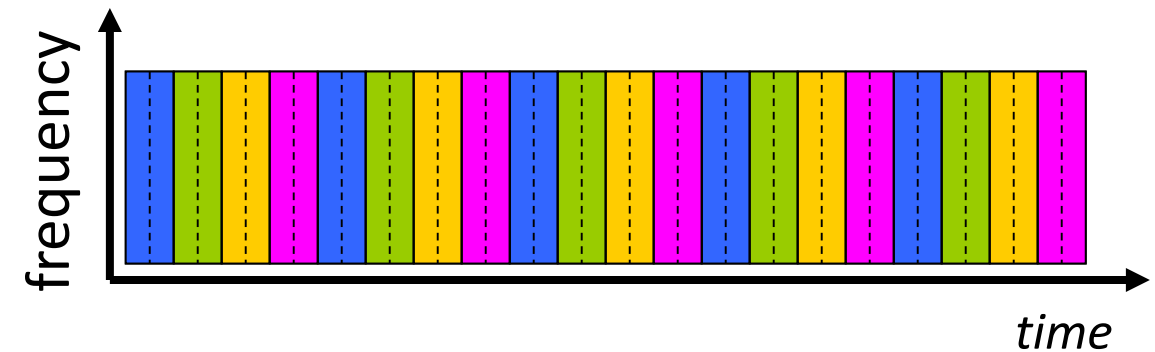
Frequency Division Multiplexing

- optical, electromagnetic frequencies divided into (narrow) frequency bands
- each call allocated its own band, can transmit at max rate of that narrow band



Time Division Multiplexing

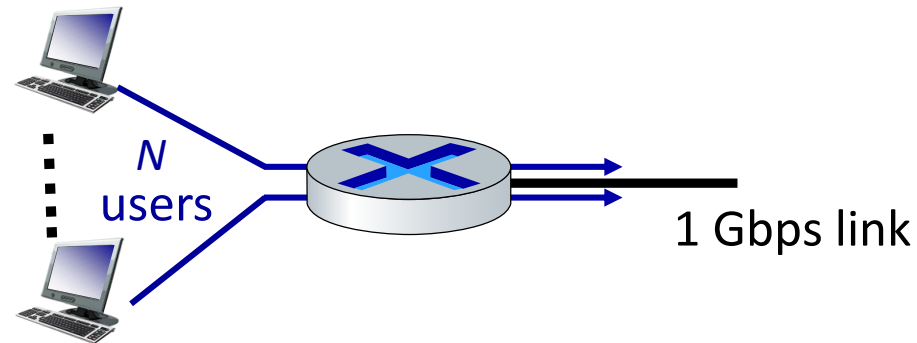
- time divided into slots
- each call allocated periodic slot(s), can transmit at maximum rate of (wider) frequency band (only) during its time slot(s)



Packet switching versus circuit switching

example:

- 1 Gb/s link
- each user:
 - 100 Mb/s when “active”
 - active 10% of time



Q: How many users can use this network under circuit-switching and packet switching?

- *circuit-switching*: 10 users
- *packet switching*: with 35 users, probability > 10 active at same time is less than .0004 *

Packet switching versus circuit switching

Is packet switching a “the perfect approach”?

- Great for “bursty” data – sometimes has data to send, but at other times not
 - resource sharing
 - simpler, no call setup
- **Excessive congestion possible: packet delay and loss** due to buffer overflow
 - protocols needed for reliable data transfer, congestion control
- **Q: How to provide circuit-like behavior with packet-switching?**
 - “It’s complicated.” We’ll study various techniques that try to make packet switching as “circuit-like” as possible.



Learning Outcomes:

- Learn the basic of networks and networking terminologies
- the types of networks and how they are designed
- The basics of Internet

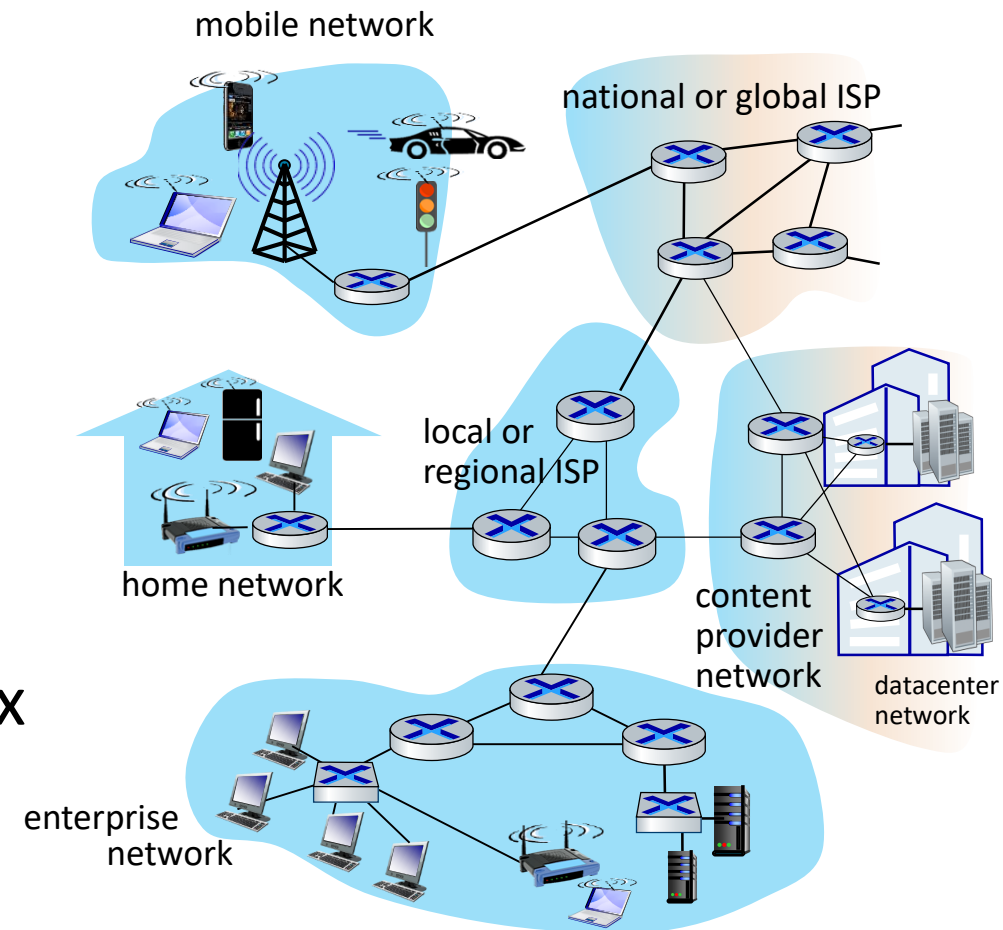


Overview/roadmap:

- We talked about
 - Data and Communication
 - Network and its types
 - Topologies
 - The Internet
 - Protocol
 - End Systems
 - Access Networks and Physical Media
 - Network Functions
- Next we discuss
 - Internet Structure

Internet structure: network of networks

- End systems connect to Internet via **access** Internet Service Providers (ISPs)
 - residential, company, university ISPs
- Access ISPs in turn must be interconnected
 - so that *any* two hosts (*anywhere!*) can send packets to each other
- Resulting network of networks is very complex
 - evolution driven by **economics and national policies**

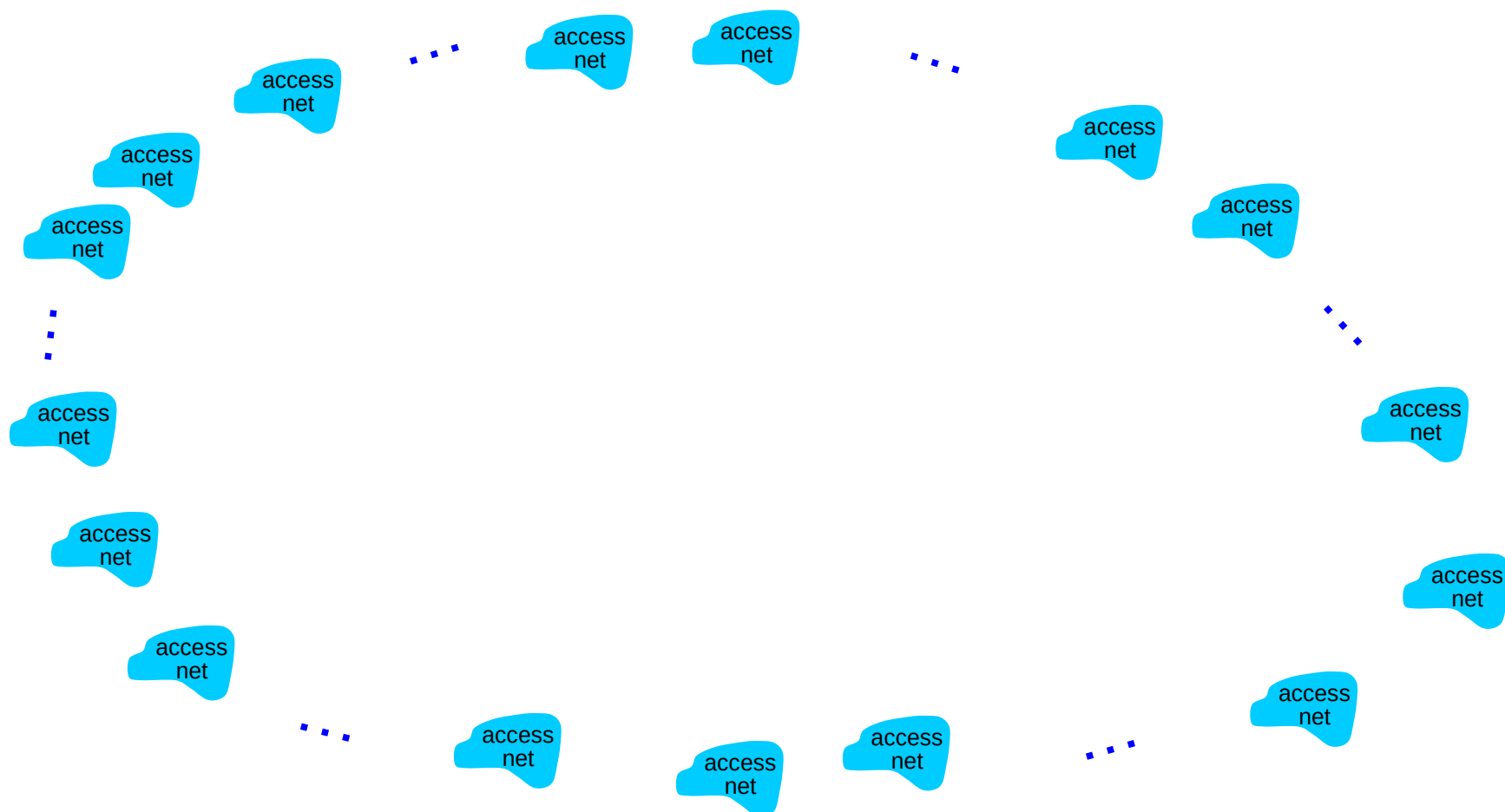


Let's take a stepwise approach to describe current Internet structure



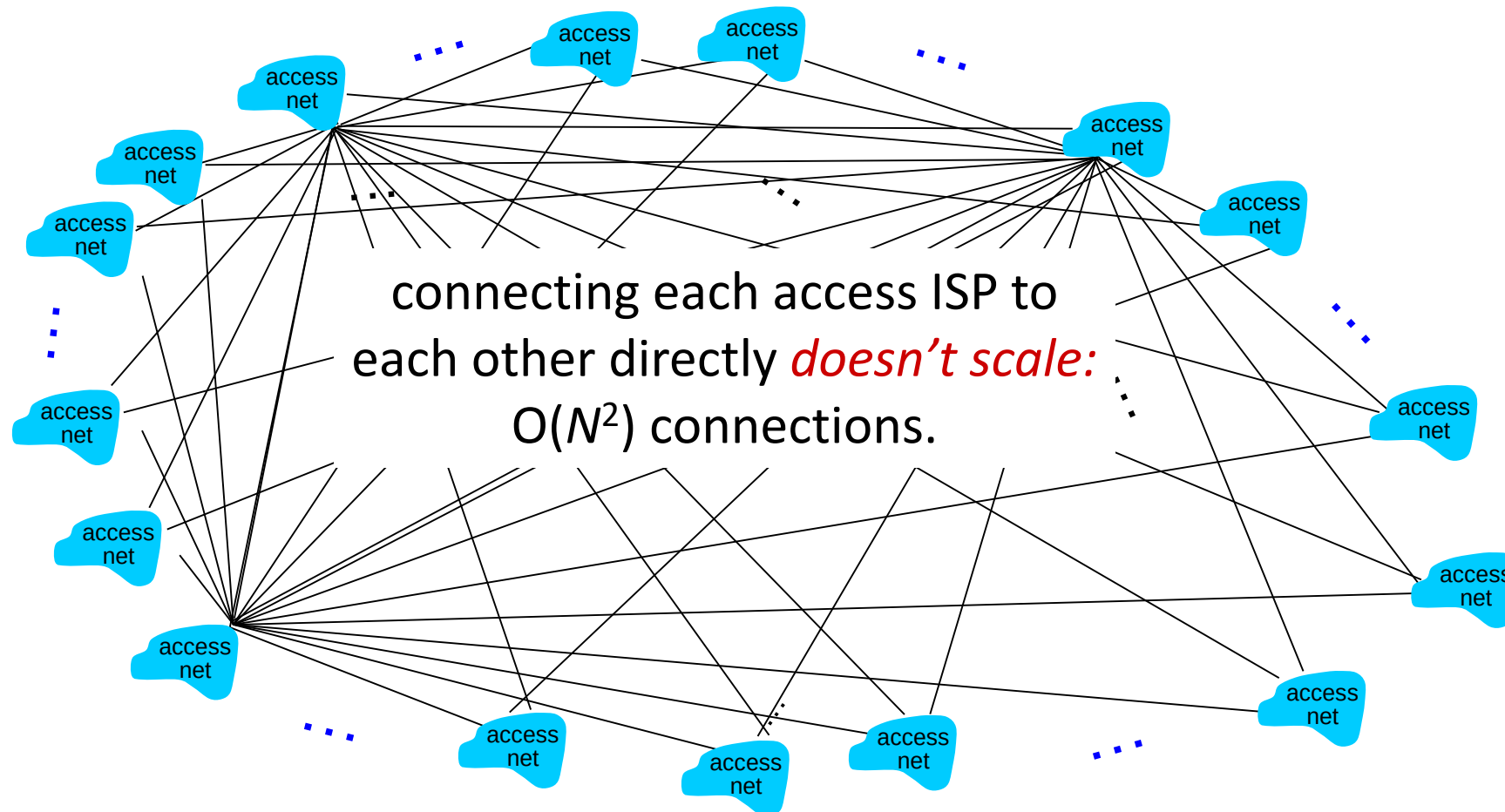
Internet structure: network of networks

Question: given *millions* of access ISPs, how to connect them together?



Internet structure: network of networks

Question: given *millions* of access ISPs, how to connect them together?

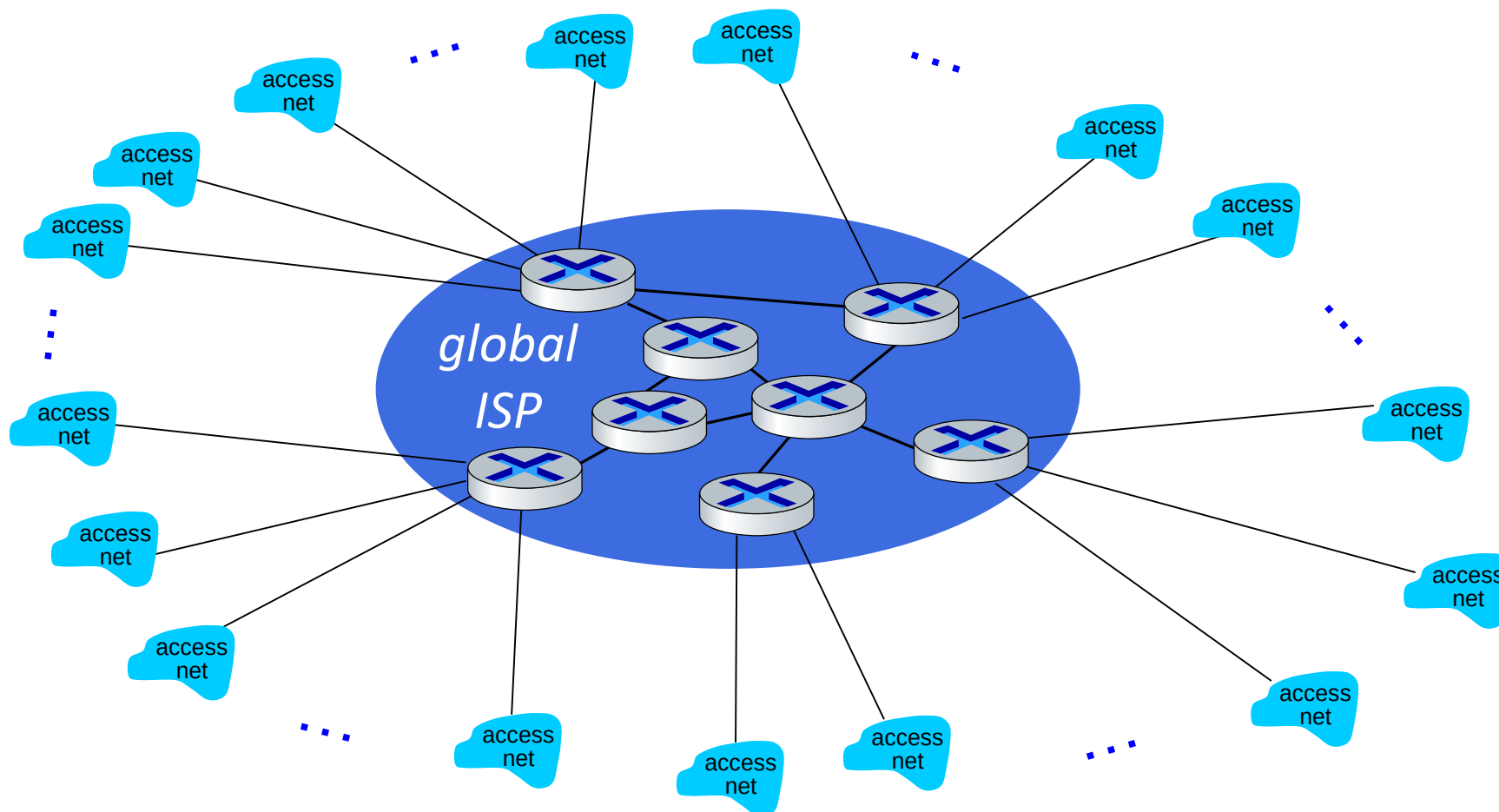




Internet structure: network of networks

Option: connect each access ISP to one global transit ISP?

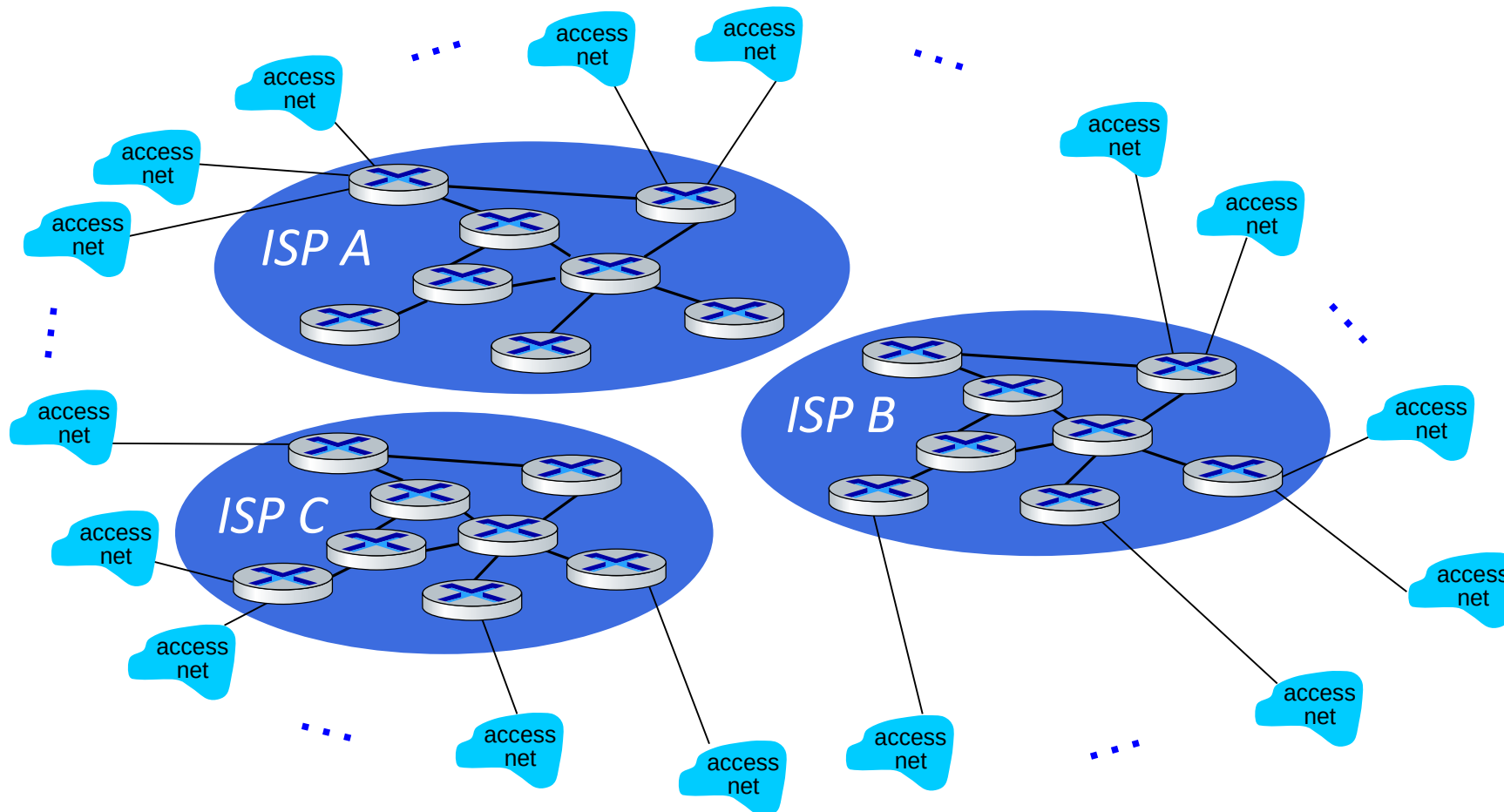
Customer and provider ISPs have economic agreement.





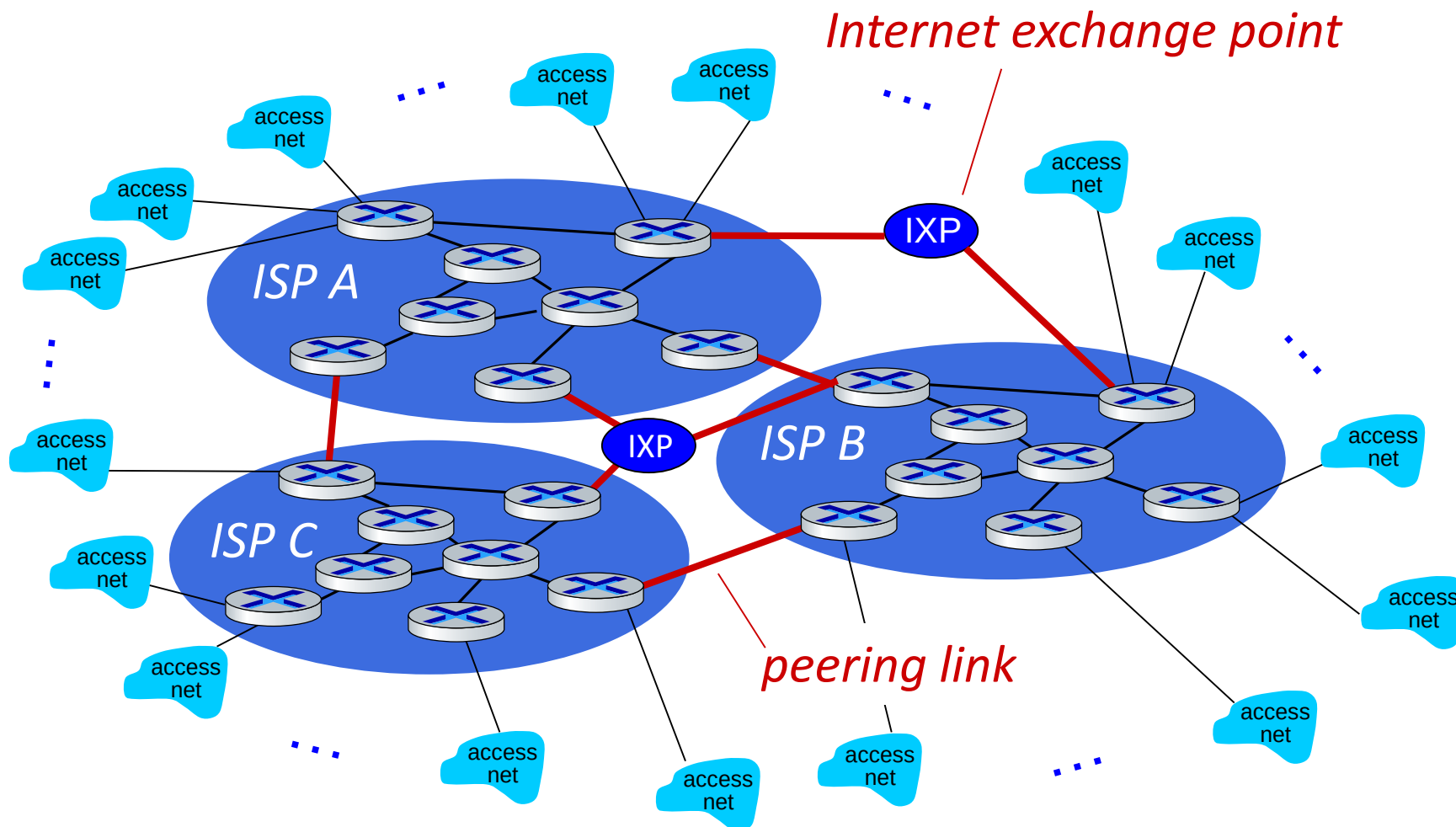
Internet structure: network of networks

But if one global ISP is viable business, there will be competitors



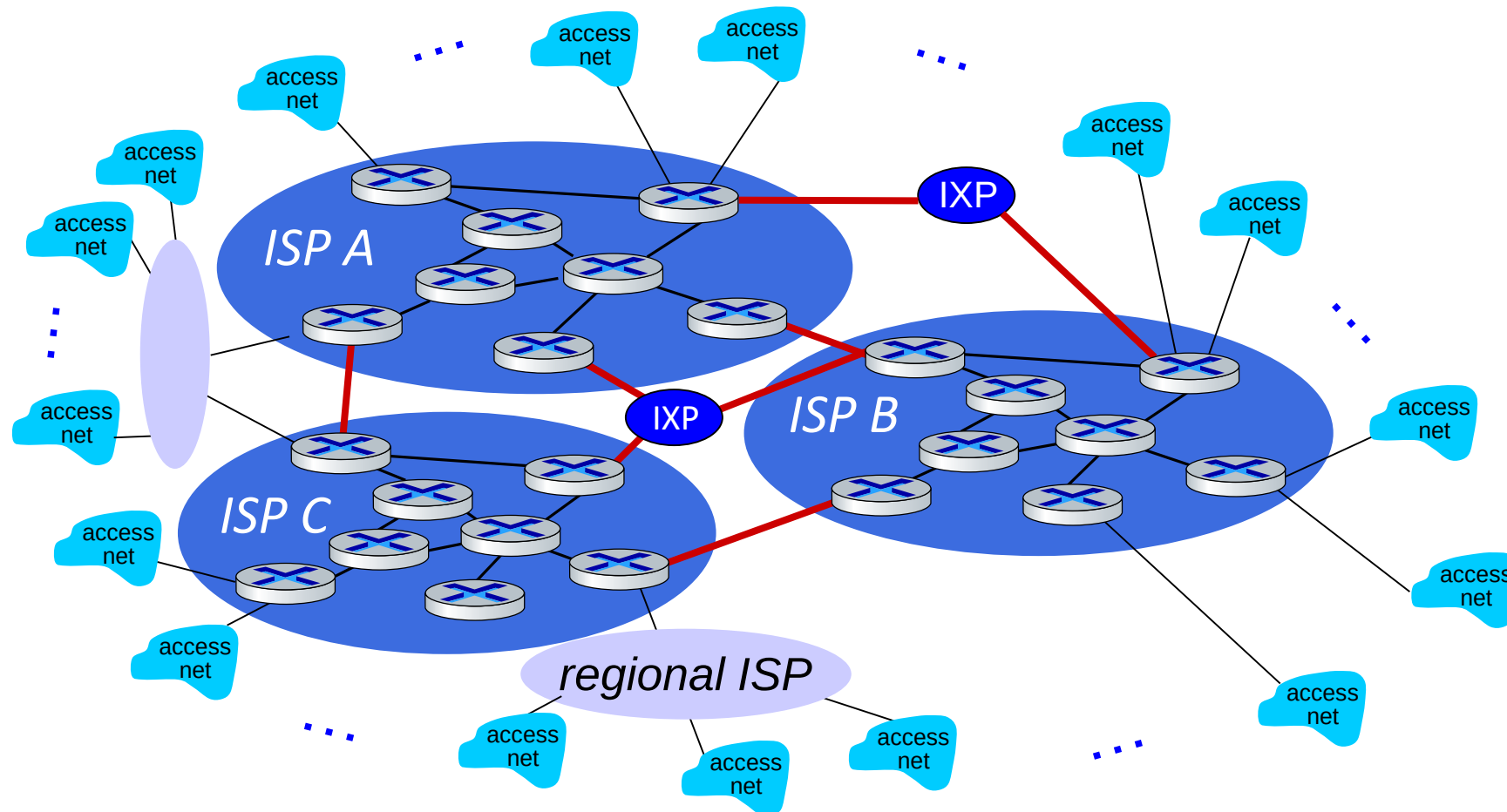
Internet structure: network of networks

But if one global ISP is viable business, there will be competitors who will want to be connected



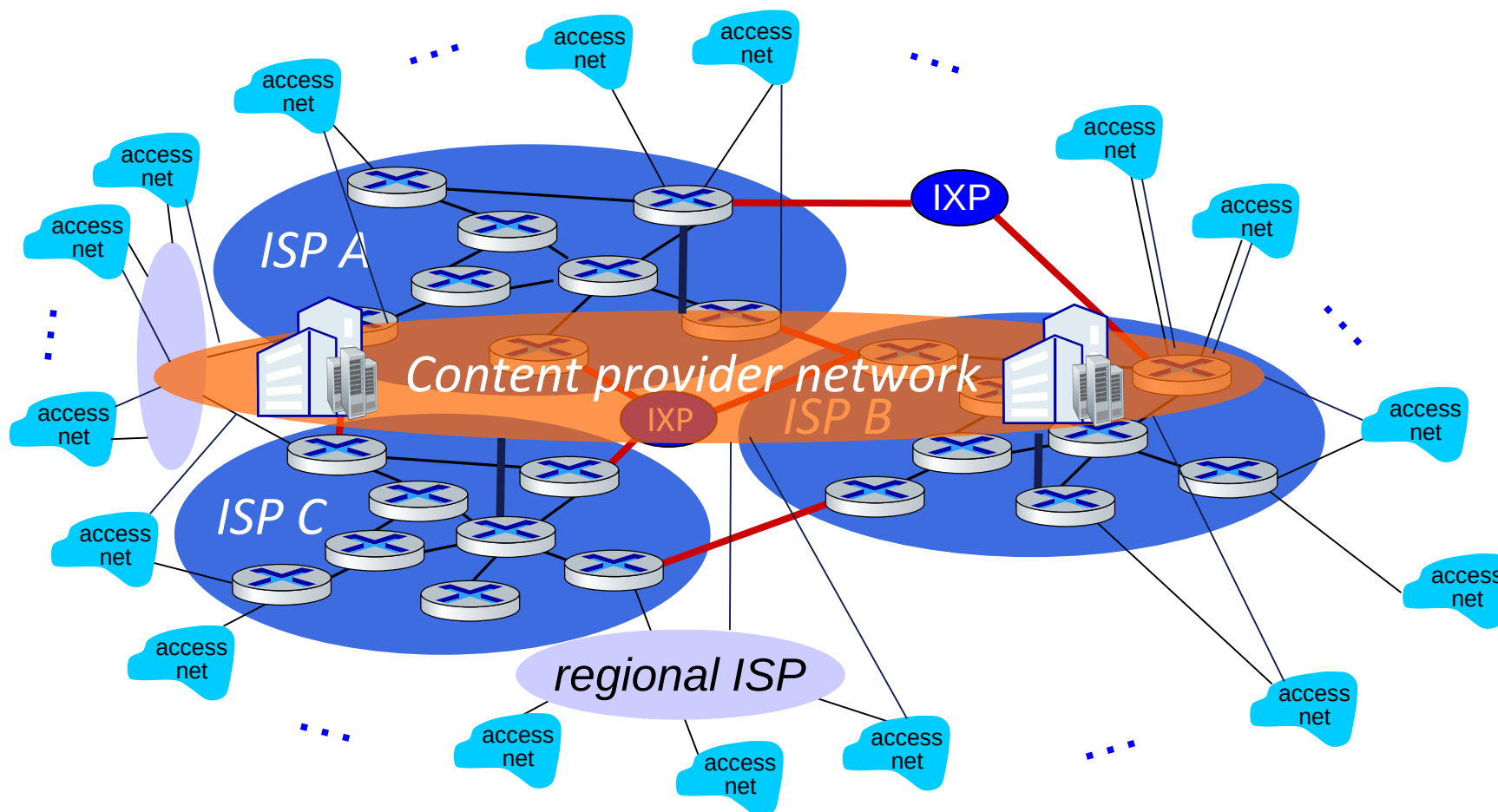
Internet structure: network of networks

... and regional networks may arise to connect access nets to ISPs

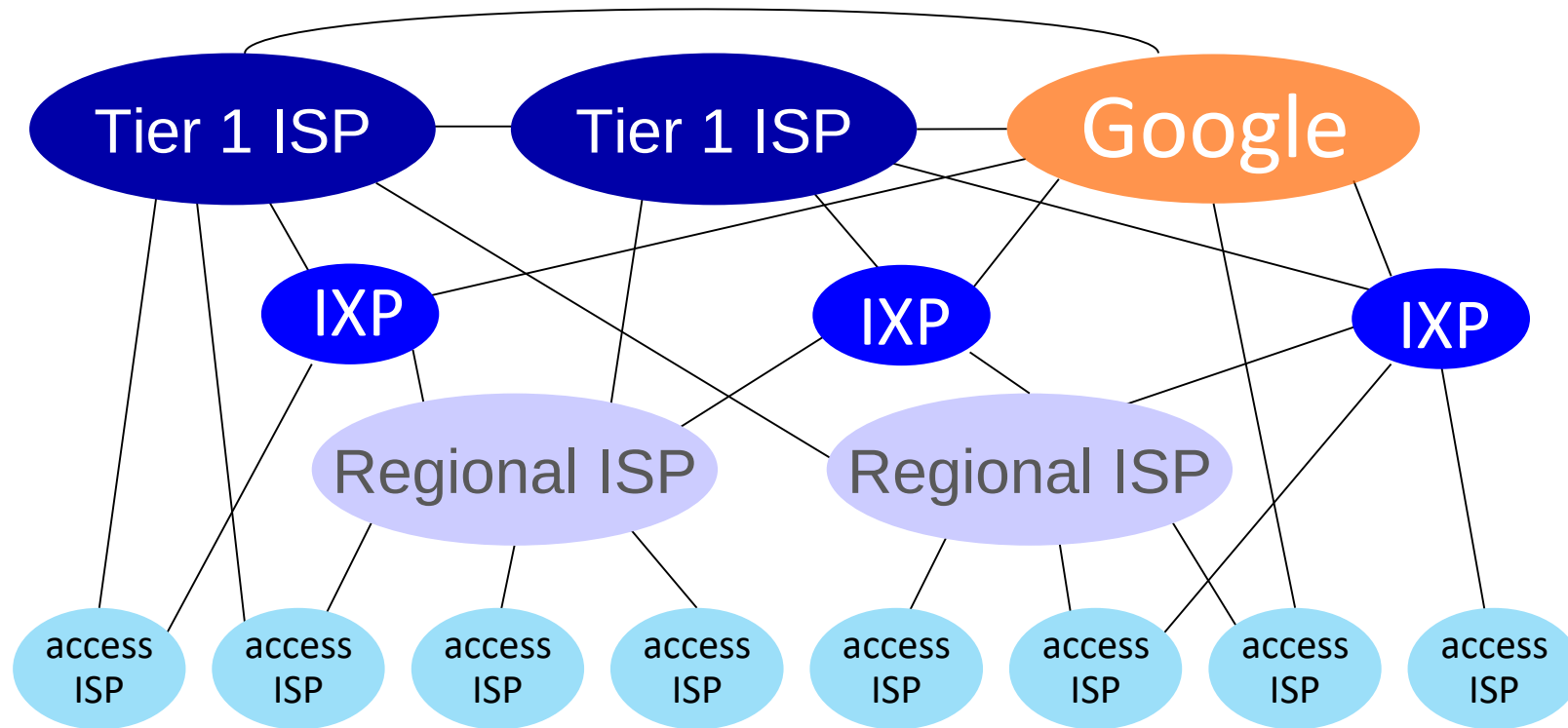


Internet structure: network of networks

... and content provider networks (e.g., Google, Microsoft, Akamai) may run their own network, to bring services, content close to end users



Internet structure: network of networks



At “center”: small # of well-connected large networks

- **“tier-1” commercial ISPs** (e.g., Level 3, Sprint, AT&T, NTT), national & international coverage
- **content provider networks** (e.g., Google, Facebook): private network that connects its data centers to Internet, often bypassing tier-1, regional ISPs



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**Thanks &
See you
soon!**